



EE115 Analog Circuits

模拟电路

Dr. Hao Ren 任豪

Office: SIST Bldg 3-328

renhao@shanghaitech.edu.cn

Outline



- Review of the course

NMOSFET (NMOS)

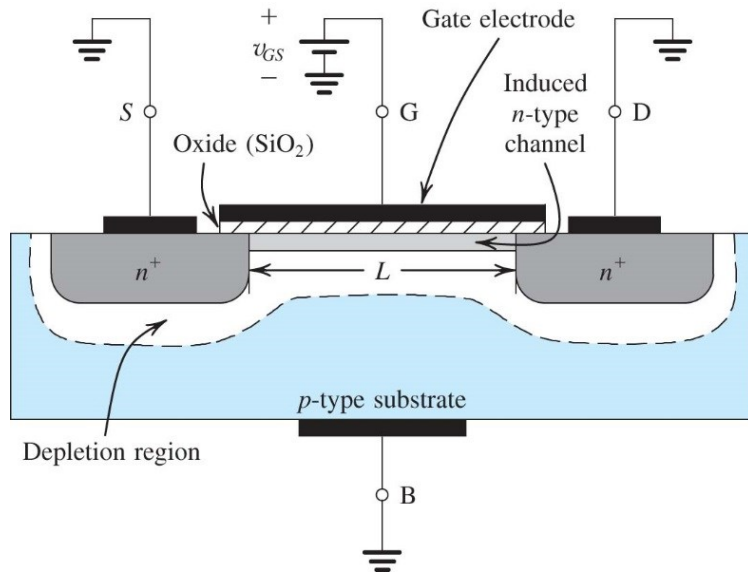


Figure 5.2 The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

- **N-channel (n-沟道) MOSFET**
 - Current conducted by e⁻s
- 3 terminal device
 - **Source (S, 源)**: n⁺ (heavily doped n-type)
 - **Drain (D, 漏)**: n⁺
 - **Gate (G, 栅)**; metal deposited on insulator above channel
- **Substrate (Body, 衬底, 体)** is a 4th terminal
 - Substrate is p-doped
- Electrons are induced in channel when a positive gate voltage is applied
- Electrons moves from Source to Drain
 - Current flows from D to S

Triode Region ($v_{DS} < v_{OV}$)

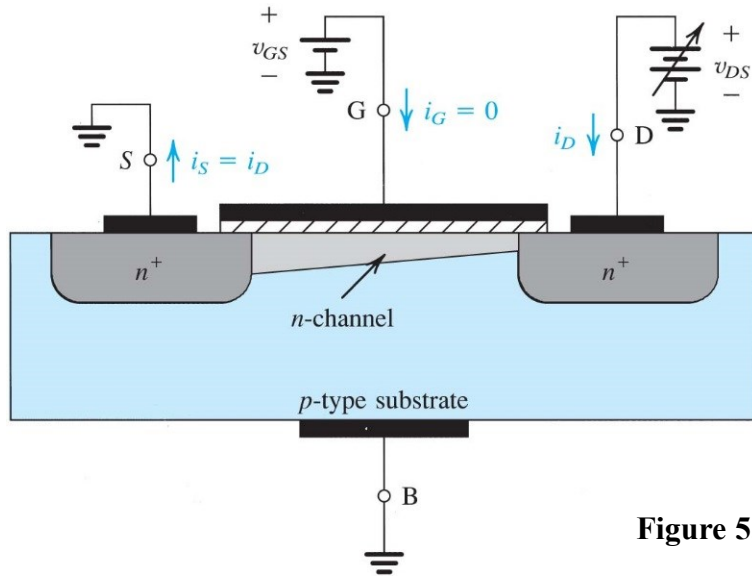
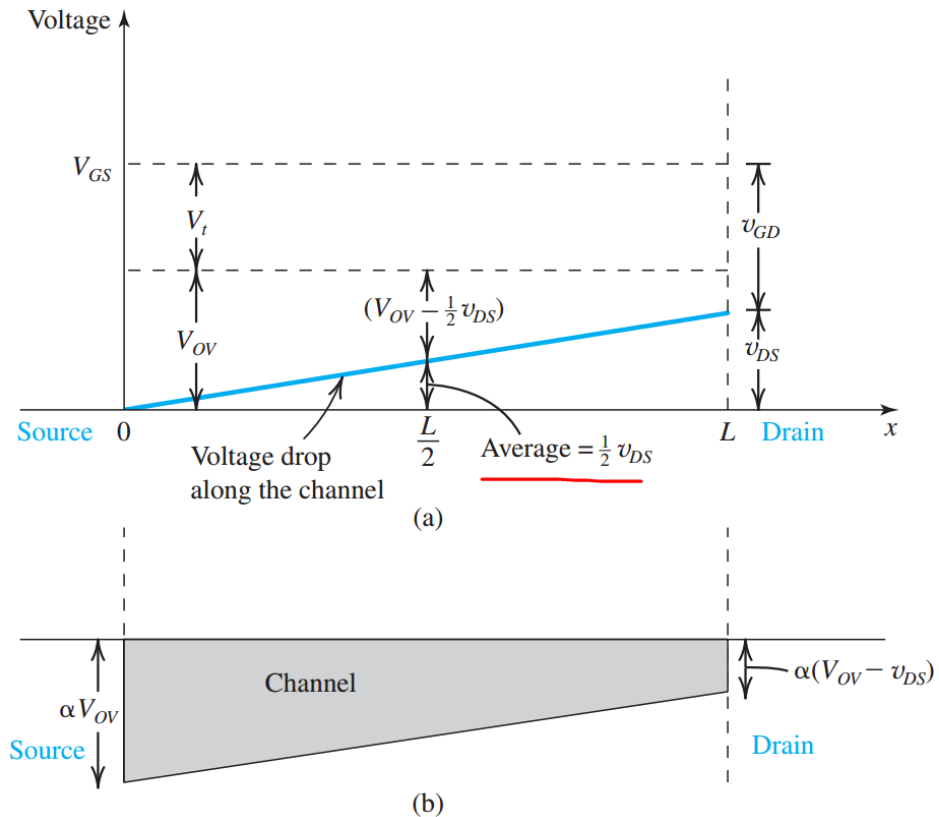


Figure 5.5

- As v_{DS} increases, the potential in the channel is no longer a constant. Assume the channel is $v(x)$:

$$i_D = k'_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} v_{DS} \right) v_{DS} = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$



I-V curve

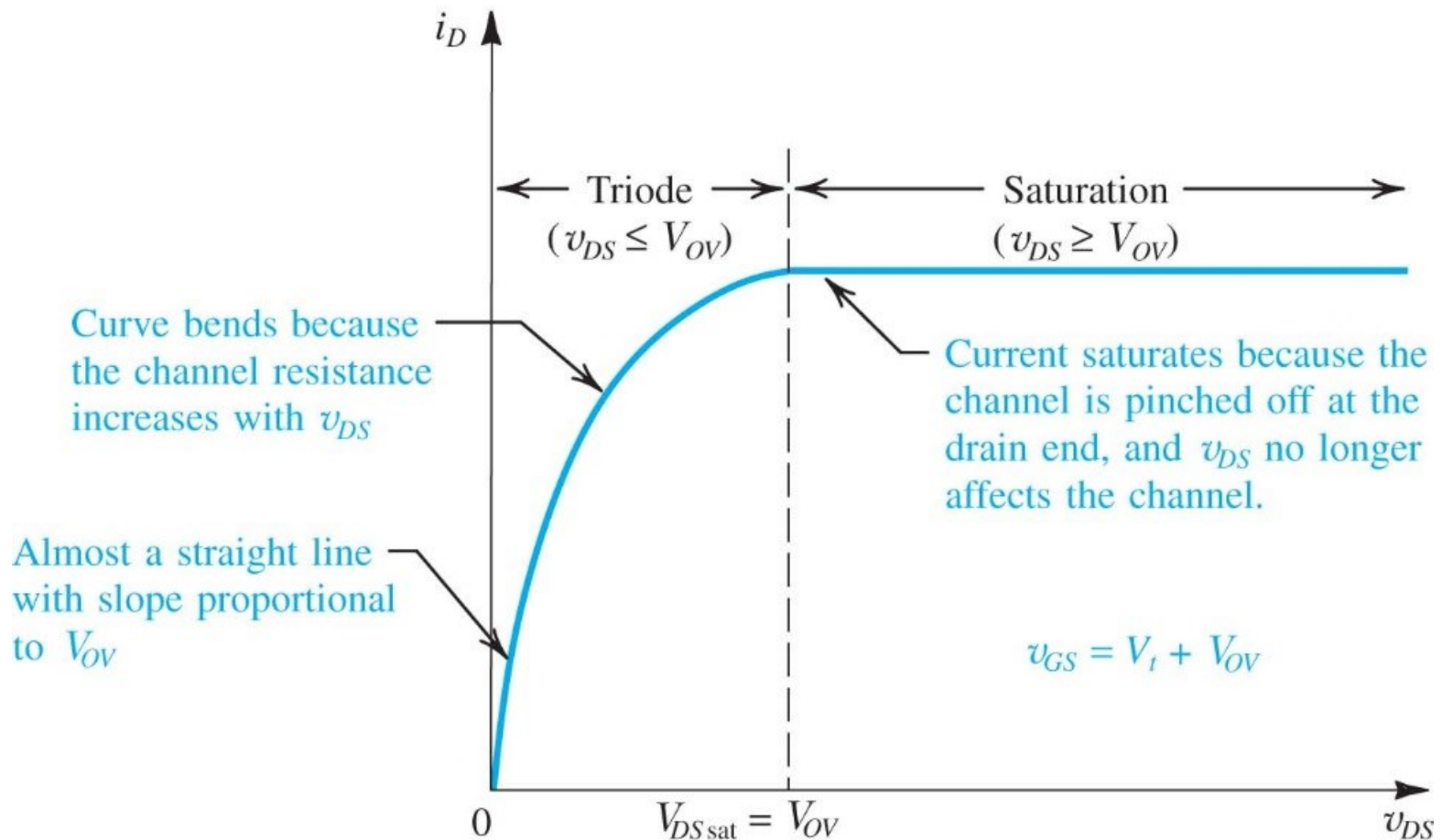


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} = V_t + V_{OV}$

Pinch-Off (夹断)

The channel potential at the drain side is v_{DS} .

When $v_{DS} = v_{OV}$, the local charge density at the drain becomes zero.

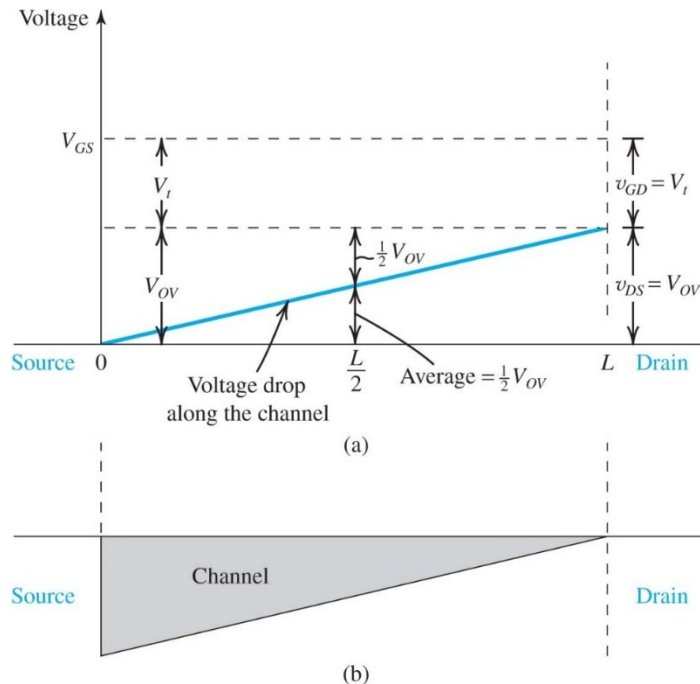


Figure 5.8

So the channel is "pinched off" near the Drain.

Once the channel is pinched off, the drain current remains constant:

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2$$

This region, $v_{DS} > v_{OV}$, is called "Saturation (饱和)"

Saturation Region ($v_{DS} \geq v_{OV}$)

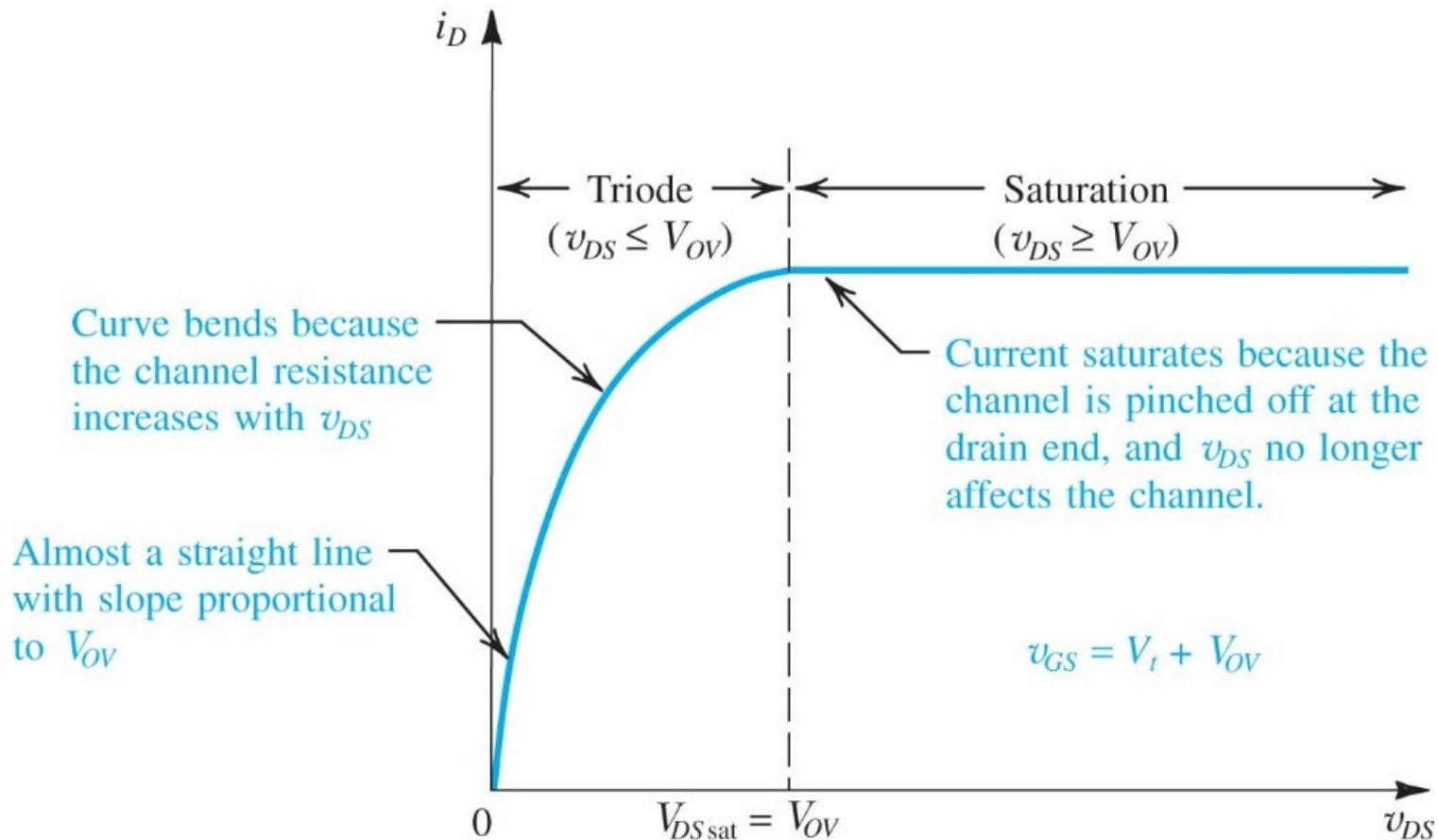


Figure 5.7 The drain current i_D versus the drain-to-source voltage v_{DS} for an enhancement-type NMOS transistor operated with $v_{GS} = V_t + V_{OV}$

I-V Curves of NMOS

$$i_D = k'_n \left(\frac{W}{L} \right) \left(V_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right)$$

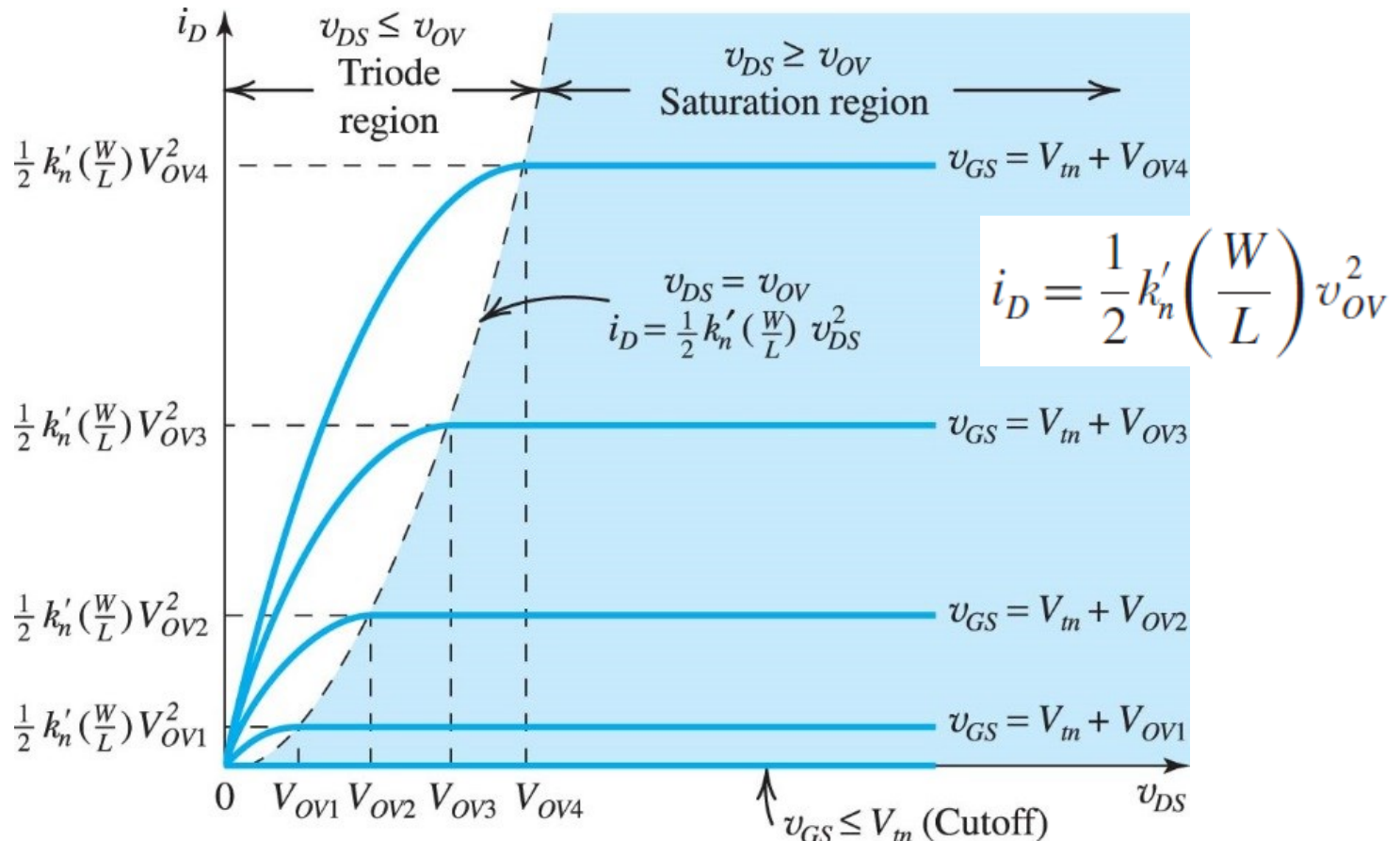
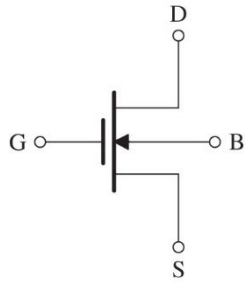
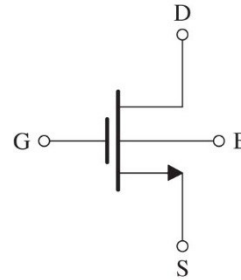


Figure 5.13 The $i_D - v_{DS}$ characteristic for an enhancement-type NMOS transistor.

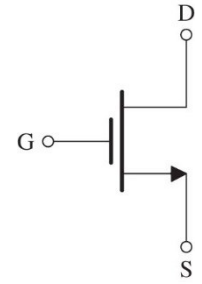
Circuit Symbol for NMOS



(a)



(b)



(c)

Figure 5.11 (a) Circuit symbol for the n -channel enhancement-type MOSFET. (b) Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., n channel). (c) Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

- 4 terminal including Body (Arrow pointing to channel indicating substrate is **p-type**)
- Modified circuit symbol with arrow on source (Arrow indicating direction of current flow)
- Simplified circuit symbol with body connected to source (or when the effect of the body on device operation is unimportant)
- Note in NMOS
 - » **Drain voltage is always more positive than source voltage**
 - » **Current always flows from Drain to Source**

Finite Output Resistance due to Channel length Modulation (沟长调制)



- When $v_{DS} = v_{OV}$
Pinched-off happens

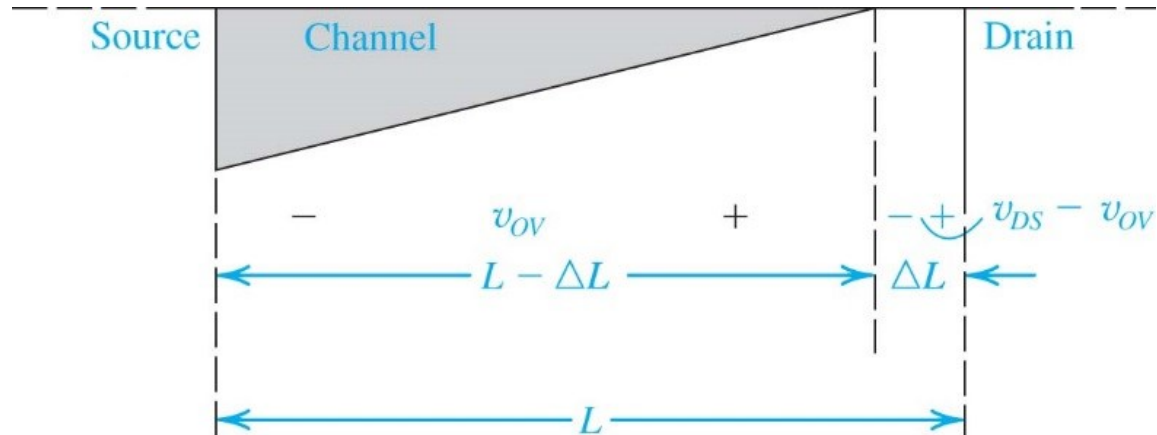


Figure 5.16 Increasing v_{DS} beyond v_{DSsat} causes the channel pinch-off point to move slightly away from the drain, thus reducing the effective channel length (by ΔL)

- When $v_{DS} > v_{OV}$
Pinch-off point shifts to the source

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_m)^2 (1 + \lambda v_{DS})$$

Output Resistance of MOSFET

■ Output resistance of NMOS:

$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}} \right]_{v_{GS} \text{ constant}}^{-1} = \left[\lambda \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_m)^2 \right]^{-1} = \frac{1}{\lambda I_D}$$

Due to CLM, the output resistance of MOSFET is not infinite, but still a large value at least for long-channel device.

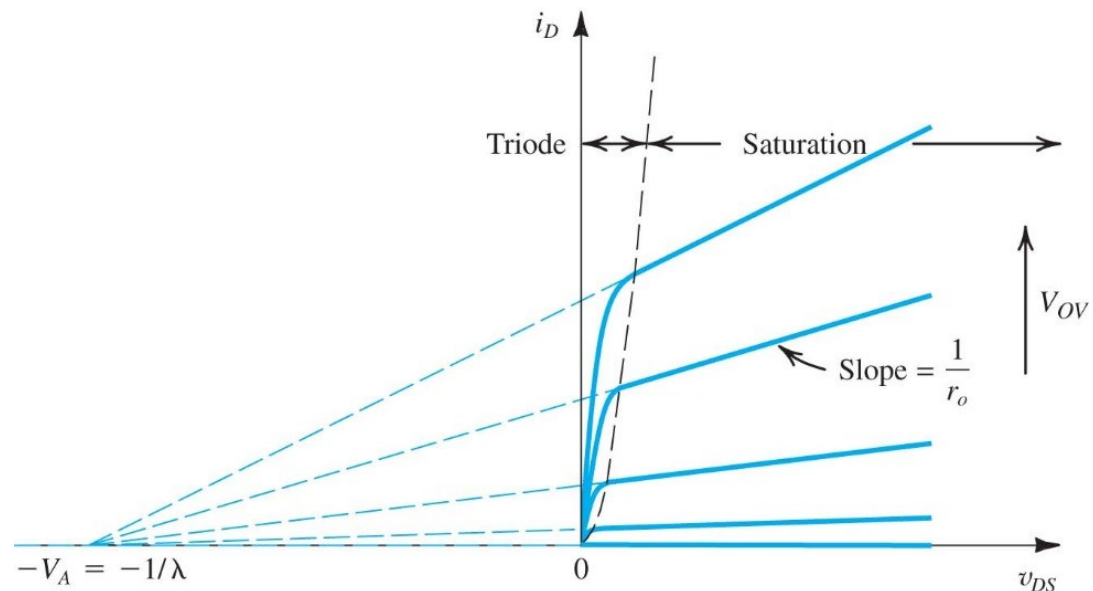
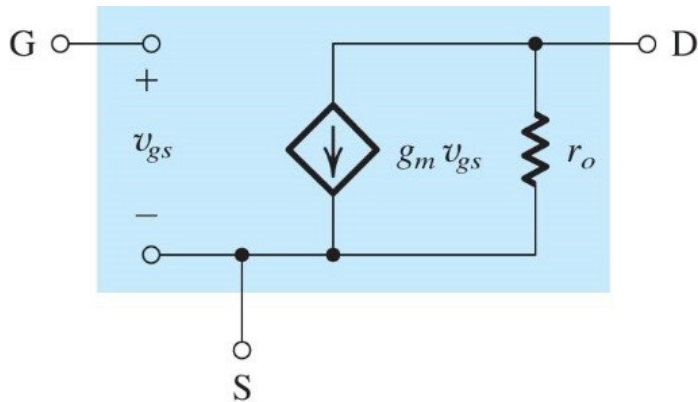


Figure 5.17 Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L .

Small signal model for MOSFET



(b)

Figure 7.13 Small-signal models for the MOSFET: (b) including the effect of channel-length modulation, modeled by output resistance $r_o = |V_A|/I_D$. These models apply equally well for both NMOS and PMOS transistors.

- The equivalent circuit is valid for both NMOS and PMOS
- In PMOS, use absolute sign for all parameters: $|V_{GS}|$, $|V_t|$, $|V_{OV}|$, $|V_A|$, and replace k_n with k_p

Transconductance:

$$g_m \equiv \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{v_{GS}=V_{GS}} \quad g_m \equiv \frac{i_d}{v_{gs}} = k_n (V_{GS} - V_t)$$

$$g_m = k_n V_{OV}$$

$$V_{OV} = \sqrt{2I_D / (k'_n (W/L))}$$

$$g_m = \sqrt{2k'_n} \sqrt{W/L} \sqrt{I_D}$$

Output resistance at drain:

$$r_o = \frac{|V_A|}{I_D}$$

$$i_D = \frac{1}{2} k_n (V_{GS} + v_{gs} - V_t)^2$$

$$= \frac{1}{2} k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t) v_{gs} + \frac{1}{2} k_n v_{gs}^2$$

$$i_d = k_n (V_{GS} - V_t) v_{gs}$$

Applying small-signal model

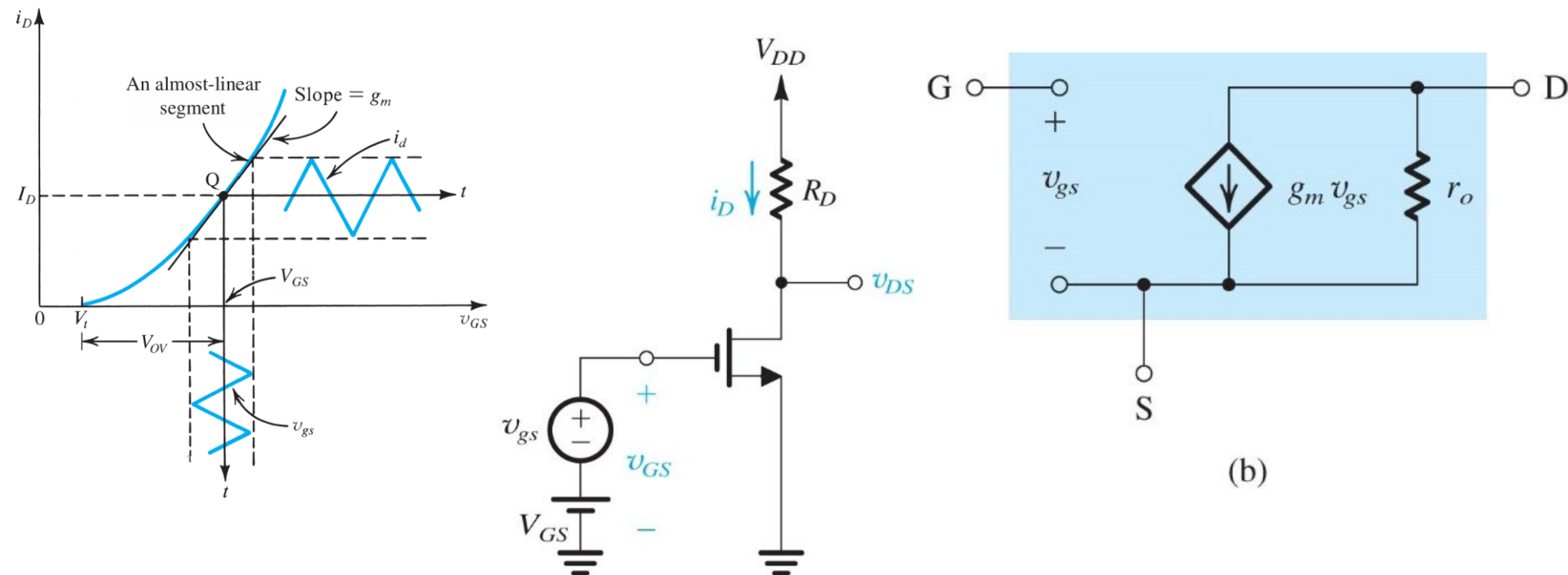


Figure 7.10 Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

$$\begin{aligned}
 i_D &= \frac{1}{2} k_n (V_{GS} + v_{gs} - V_t)^2 \\
 &= \frac{1}{2} k_n (V_{GS} - V_t)^2 + k_n (V_{GS} - V_t) v_{gs} + \frac{1}{2} k_n v_{gs}^2
 \end{aligned}$$

Systematic Procedure for Transistor Amplifier Analysis



- 1. Perform **DC analysis** (ignore small signal source)
- 2. Calculate **small-signal parameters** (g_m , r_o , etc)
- 3. Generate **AC small-signal equivalent circuit**
 - » Replace DC voltage source by short circuit
 - » Replace DC current source by open circuit
 - » Replace transistor by hybrid- π model (or other model)
- 4. Perform **circuit analysis** to determine voltage gain or other amplifier performance parameters

Common-Source (CS) Amplifier

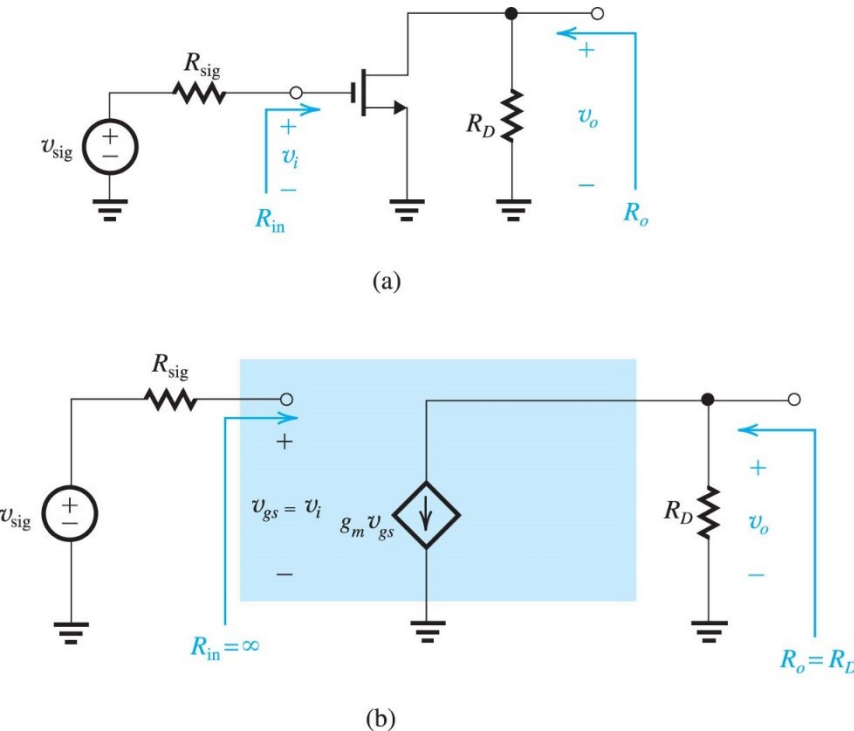


Figure 7.35 (a) Common-source amplifier fed with a signal v_{sig} from a generator with a resistance R_{sig} . The bias circuit is omitted. (b) The common-source amplifier with the MOSFET replaced with its hybrid- π model.

- Previously, using MOSFET equation, we derived analytically :

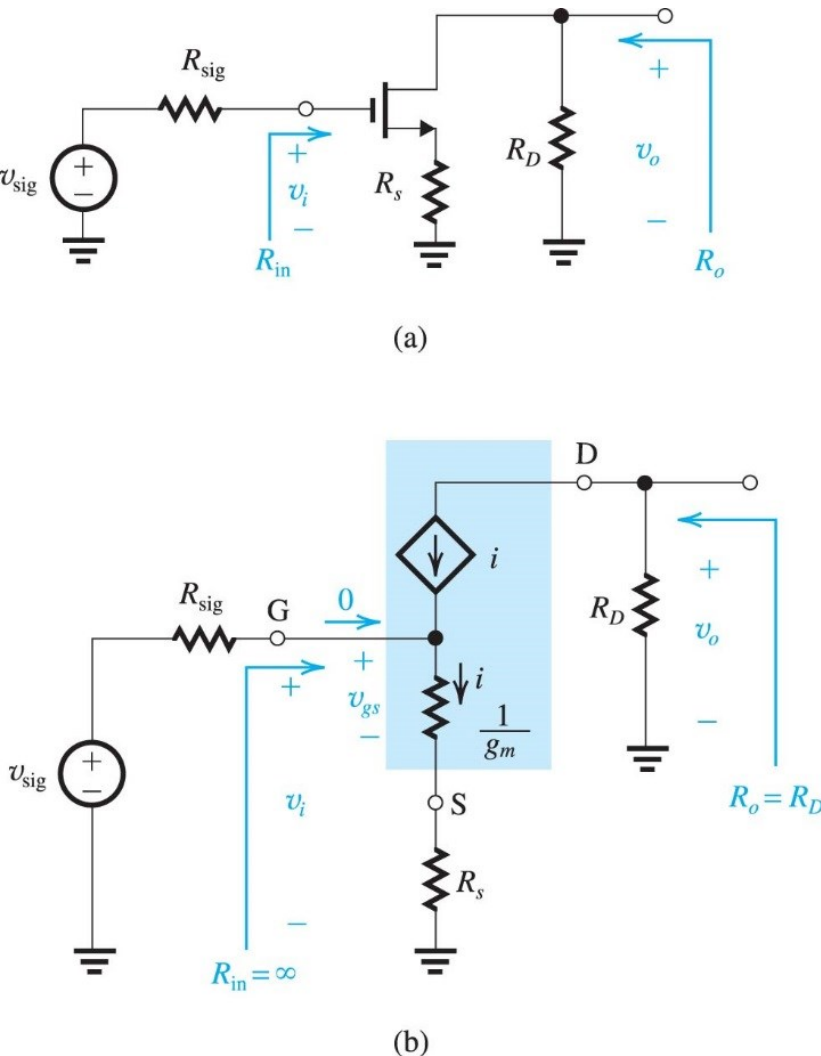
$$A_V = -k_n V_{OV} R_D$$
(The process is cumbersome, tedious, and will go quickly out of hand with more complex circuits)
- Using hybrid-pi model, it is almost trivial to solve: $A_V = -g_m R_D$
- With load resistance, R_L :
Since R_L is in parallel with R_D

$$G_V = -g_m (R_D || R_L)$$

$$R_{in} = \infty$$

$$R_o = R_D$$

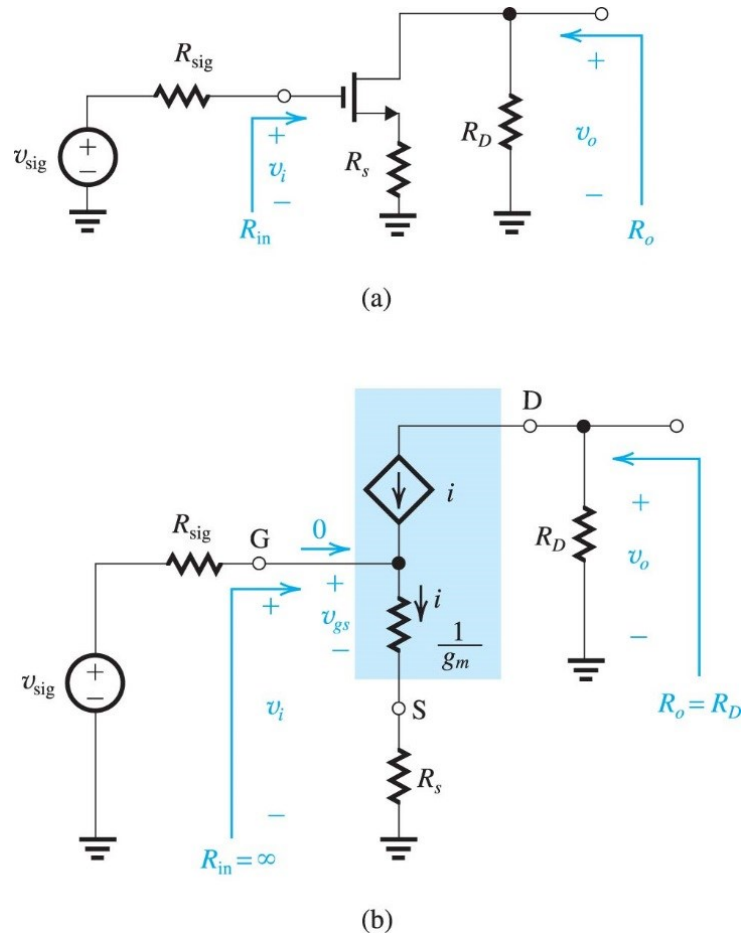
CS with Source Resistance (Source Degeneration)



- You can analyze the circuit using hybrid-pi model. (Try it !)
- However, whenever **there is a resistor connected to source**, it is much easier to use the "T-model"

Figure 7.37 The CS amplifier with a source resistance R_s : (a) circuit without bias details; (b) equivalent circuit with the MOSFET represented by its T model.

Continued



• Note

$$A_v = -\frac{R_D}{\frac{1}{g_m} + R_s} = -\frac{\text{Total resistance in Drain}}{\text{Total resistance in Source}}$$

• R_s provides negative feedback, which

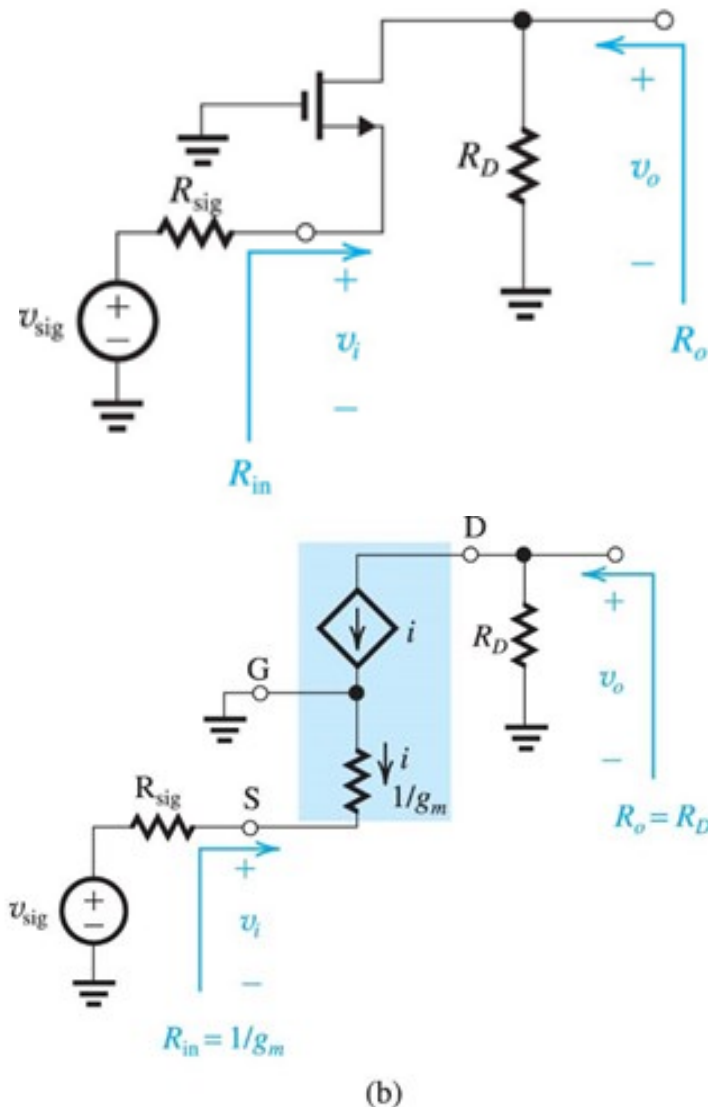
- Stabilize drain current
- Increase linearity by keeping v_{gs} small

$$v_{gs} = v_i \frac{1/g_m}{1/g_m + R_s} = v_i \frac{g_m}{1 + g_m R_s}$$

- Increase usable bandwidth

Figure 7.37 The CS amplifier with a source resistance R_s : **(a)** circuit without bias details; **(b)** equivalent circuit with the MOSFET represented by its T model.

Common-Gate (CG) Amplifier



R_{sig} connected to source
→ Use the "T-model"

$$R_{in} = \frac{1}{g_m}$$

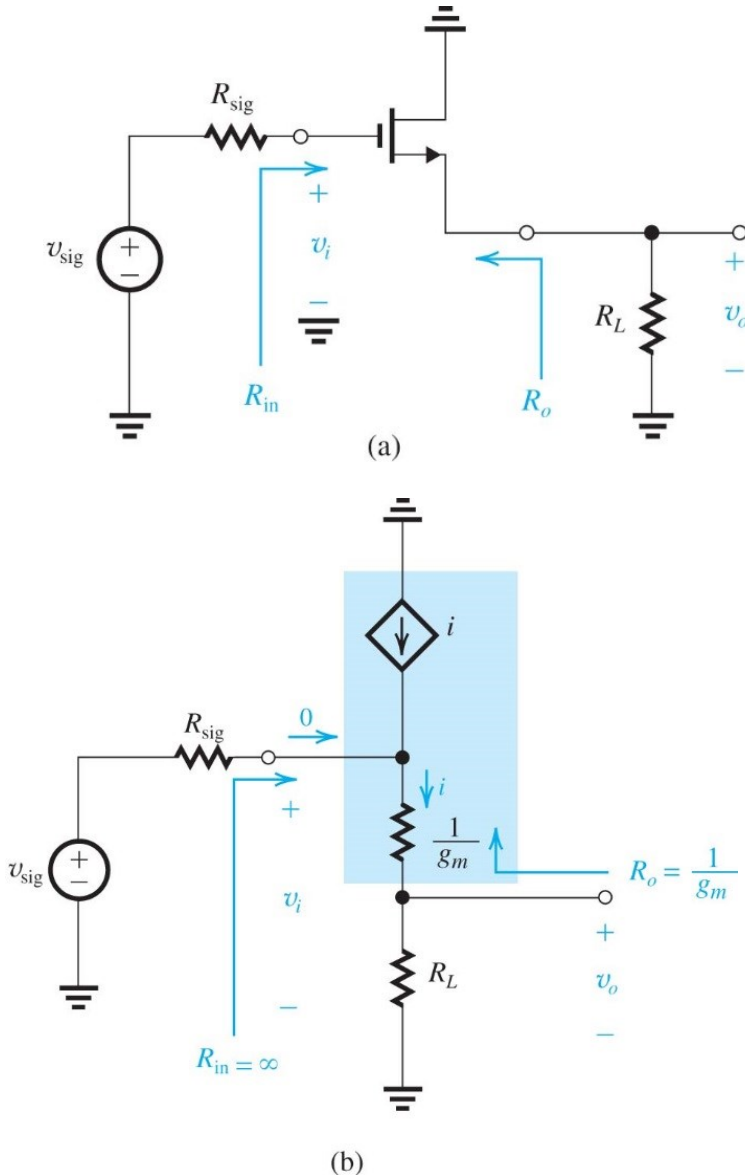
$$A_{vo} \equiv \frac{v_o}{v_i} = g_m R_D$$

$$R_o = R_D$$

$$G_v = \frac{1/g_m}{R_{sig} + 1/g_m} [g_m (R_D \parallel R_L)]$$

Figure 7.39 (a) Common-gate (CG) amplifier with bias arrangement omitted. (b) Equivalent circuit of the CG amplifier with the MOSFET replaced with its T model.

Common-Drain (CD) Amplifier (Source Follower)



R_L connected to source
→ Use the "T-model"

$$R_{in} = \infty$$

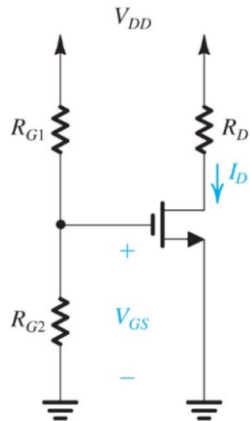
$$A_v \equiv \frac{v_o}{v_i} = \frac{R_L}{R_L + 1/g_m}$$

$$R_o = 1/g_m$$

$$G_v = A_v = \frac{R_L}{R_L + 1/g_m}$$

Figure 7.42 (a) Common-drain amplifier or source follower with the bias circuit omitted. (b) Equivalent circuit of the source follower obtained by replacing the MOSFET with its T model.

Current mirror

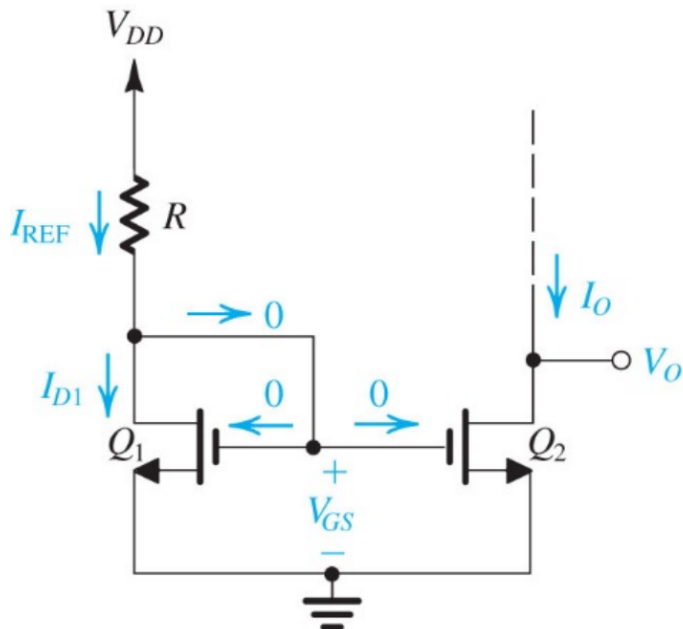


- Neglecting channel length modulation

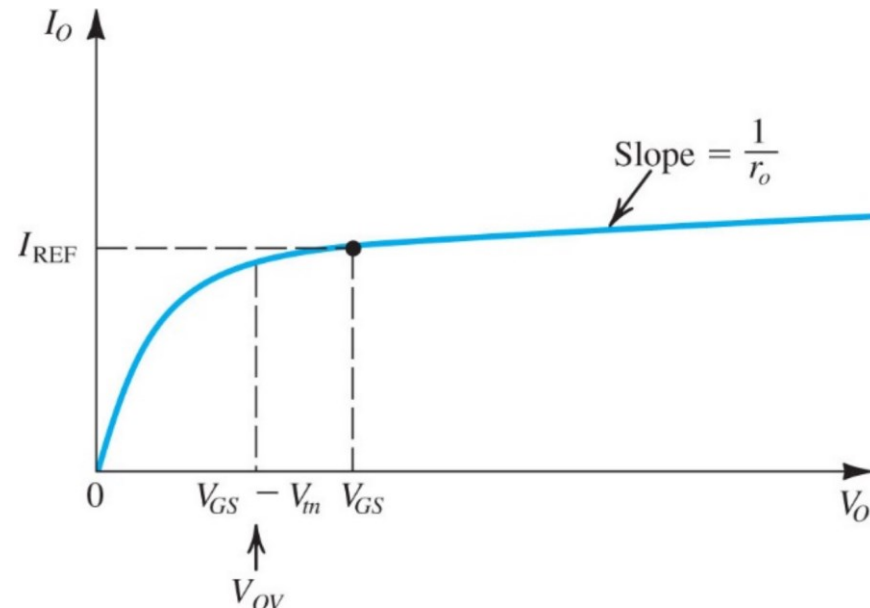
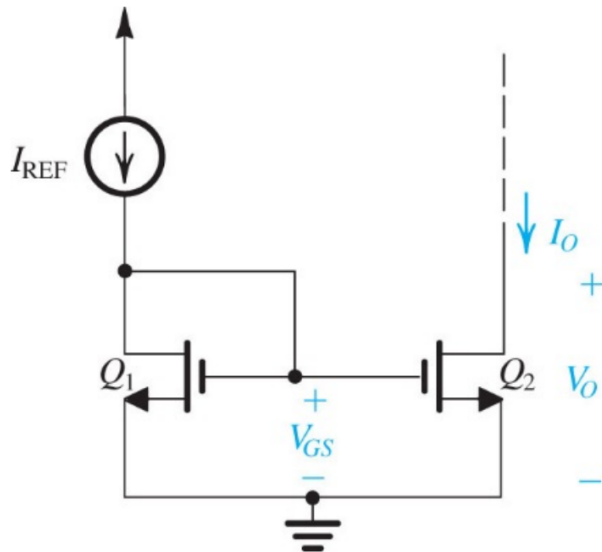
$$I_{D1} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_1 (V_{GS} - V_{tn})^2$$

$$I_O = I_{D2} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_2 (V_{GS} - V_{tn})^2$$

$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$



Current Mirrors with reference source



- Neglecting channel length modulation

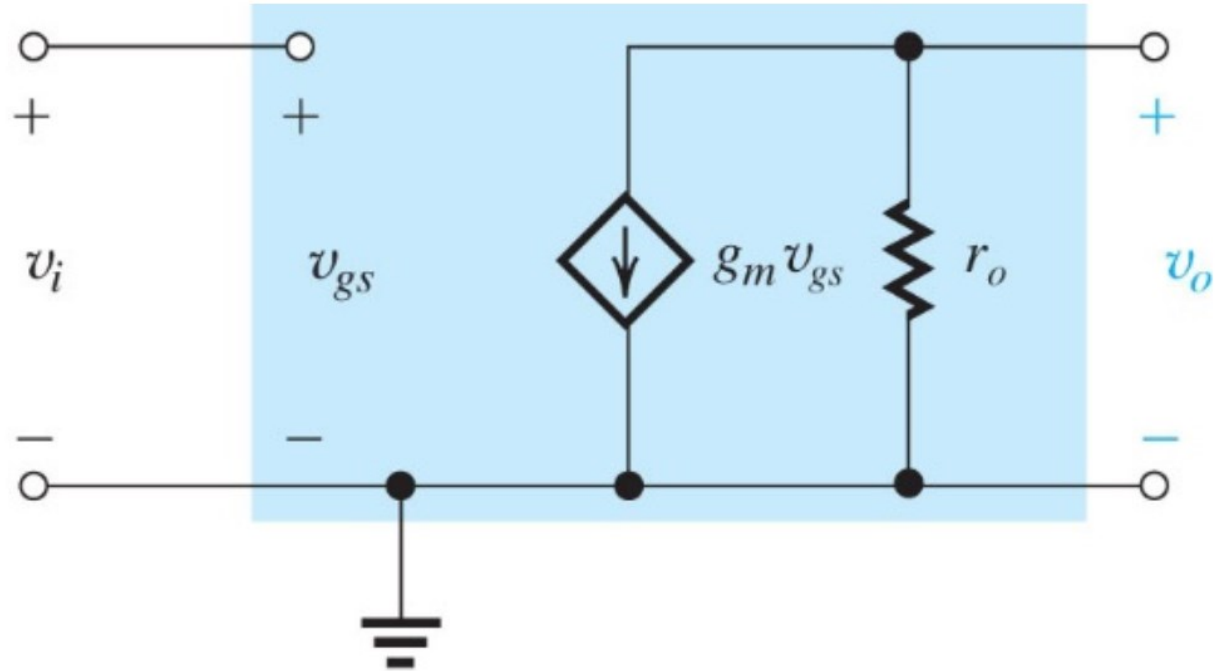
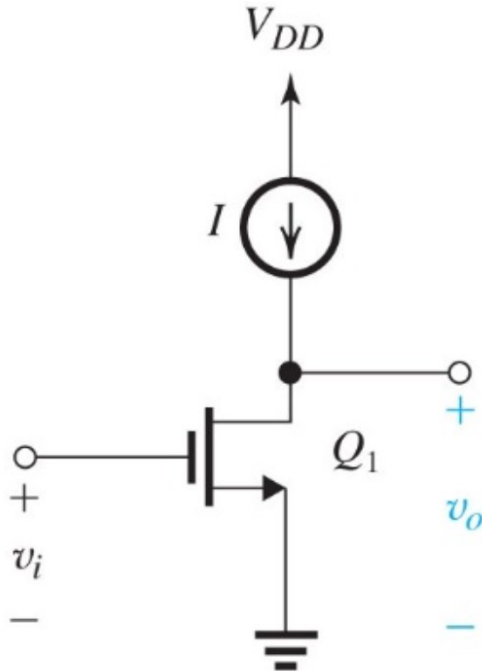
$$\frac{I_O}{I_{\text{REF}}} = \frac{(W/L)_2}{(W/L)_1}$$

- If considering channel length modulation

$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{\text{REF}} \left(1 + \frac{V_O - V_{GS}}{V_{A2}} \right)$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_m)^2 (1 + \lambda v_{DS})$$

Basic Gain Cells with Ideal Current Source Load



$$g_m = \frac{I_D}{V_{OV}/2}$$

$$g_m = \sqrt{2\mu_n C_{ox}(W/L)}\sqrt{I_D}$$

$$r_o = \frac{V_A}{I_D} = \frac{V'_A L}{I_D}$$

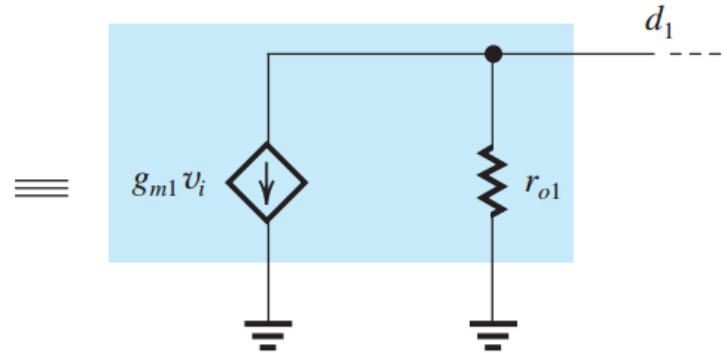
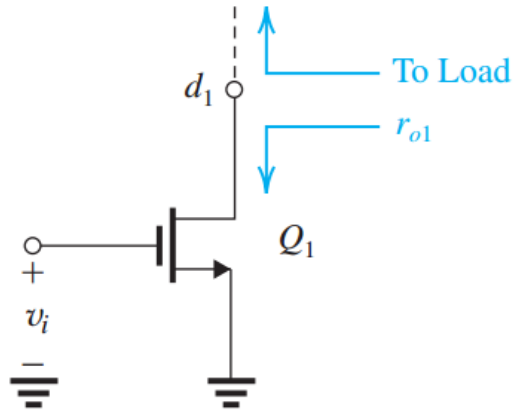
$$A_0 = \frac{V_A}{V_{OV}/2} = \frac{2V'_A L}{V_{OV}}$$

$$R_{in} = \infty$$

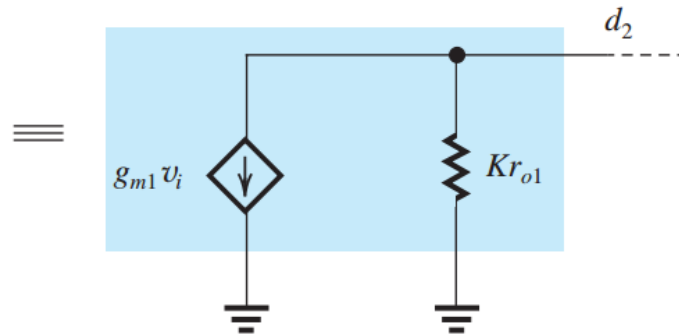
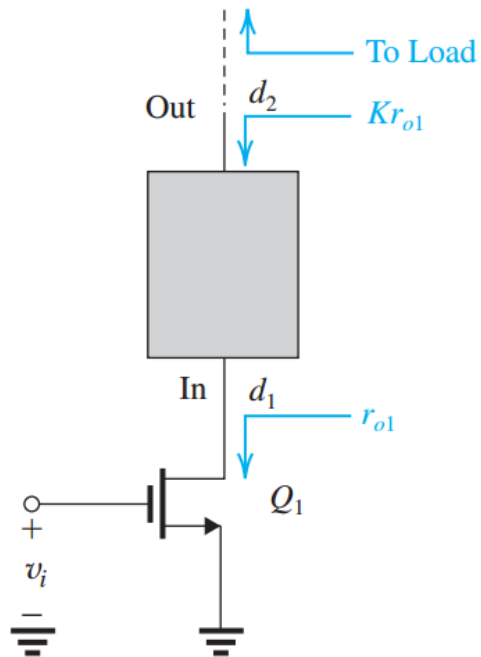
Intrinsic gain: $A_{vo} = -g_m r_o$

$$R_o = r_o$$

How To Increase Voltage Gain?

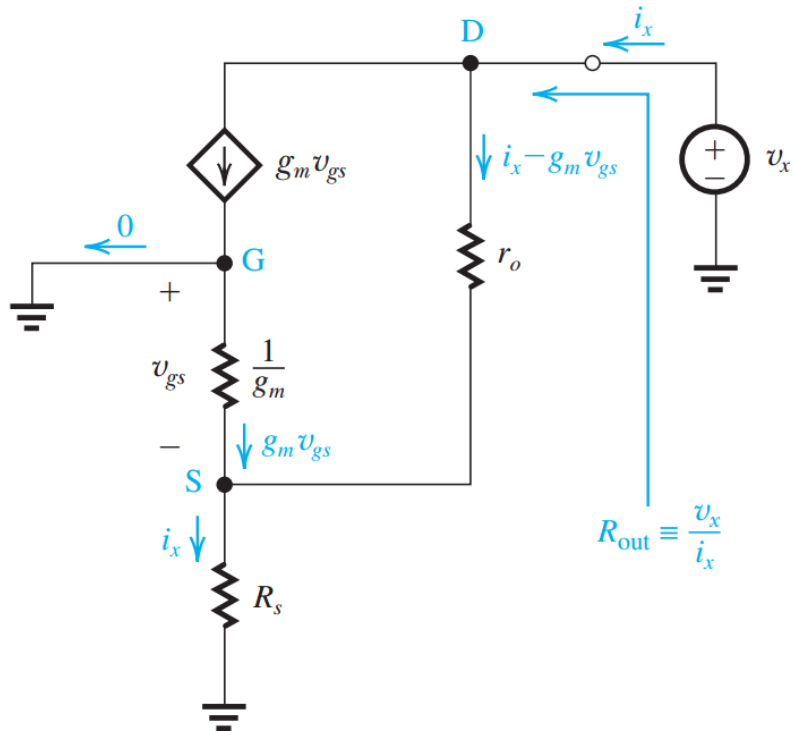


(a)



(b)

Continued 1



$$v_x = (i_x - g_m v_{gs})r_o + i_x R_s$$

$$v_{gs} = -i_x R_s$$

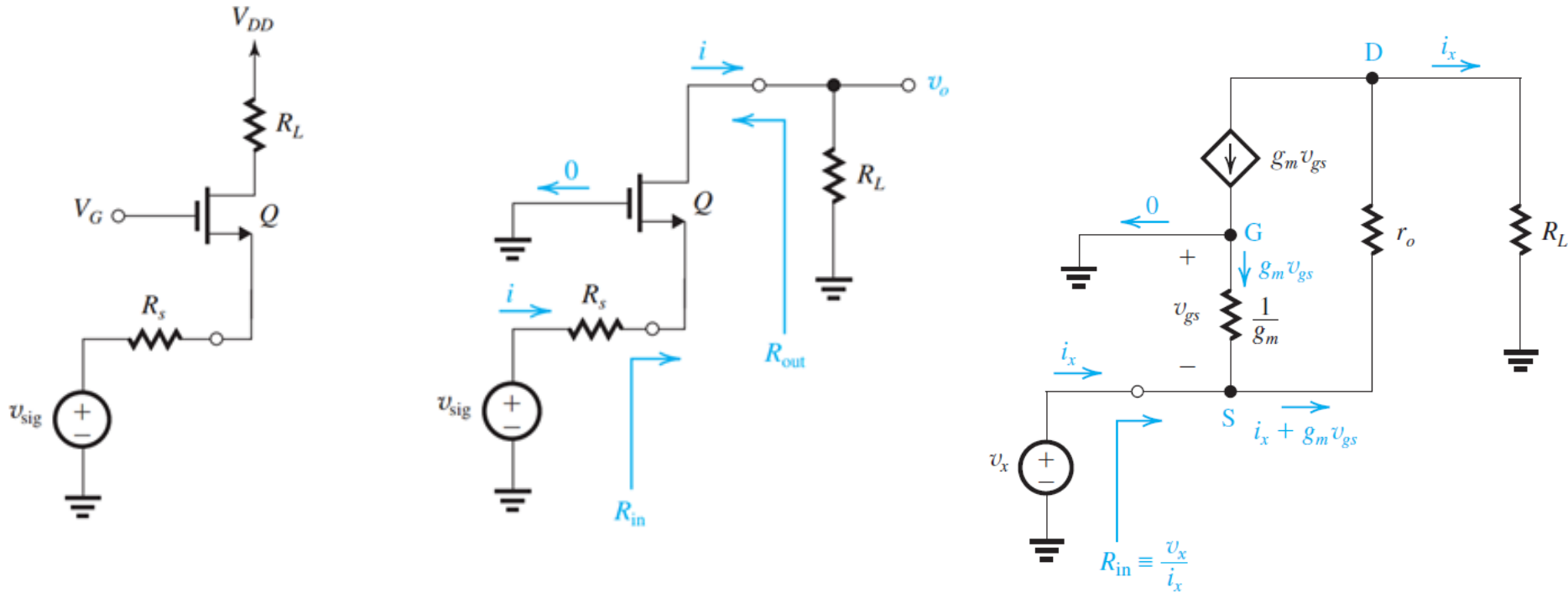
$$R_{\text{out}} = r_o + R_s + g_m r_o R_s$$

$$= r_o + (1 + g_m r_o) R_s$$

$$\simeq (g_m r_o) R_s$$

The source impedance is transformed to the output by multiplying the intrinsic gain.

Impedance Transformation of Common Gate Amplifier

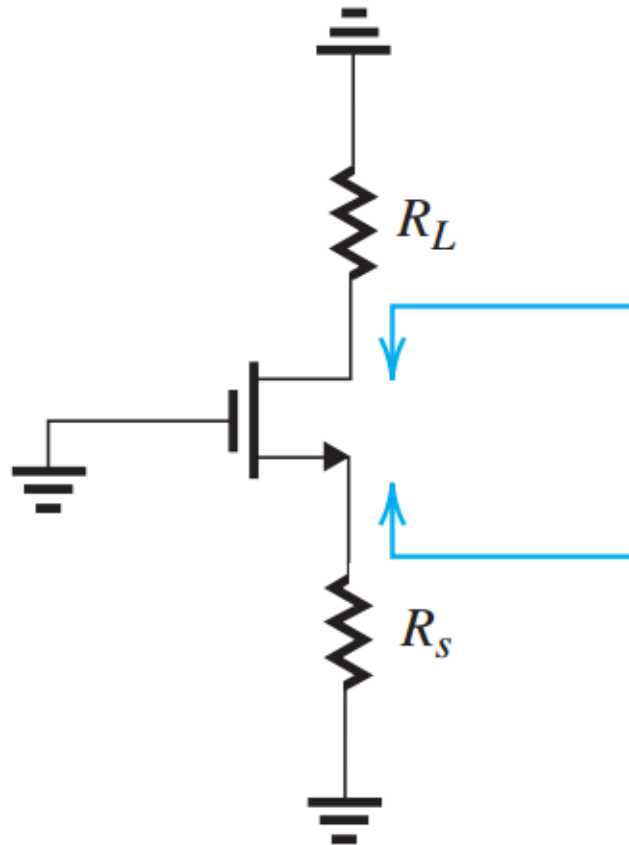


$$v_x = (i_x + g_m v_{gs}) r_o + i_x R_L$$

$$R_{in} = \frac{r_o + R_L}{1 + g_m r_o} \simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

The load impedance is transformed to the input by dividing by the intrinsic gain.

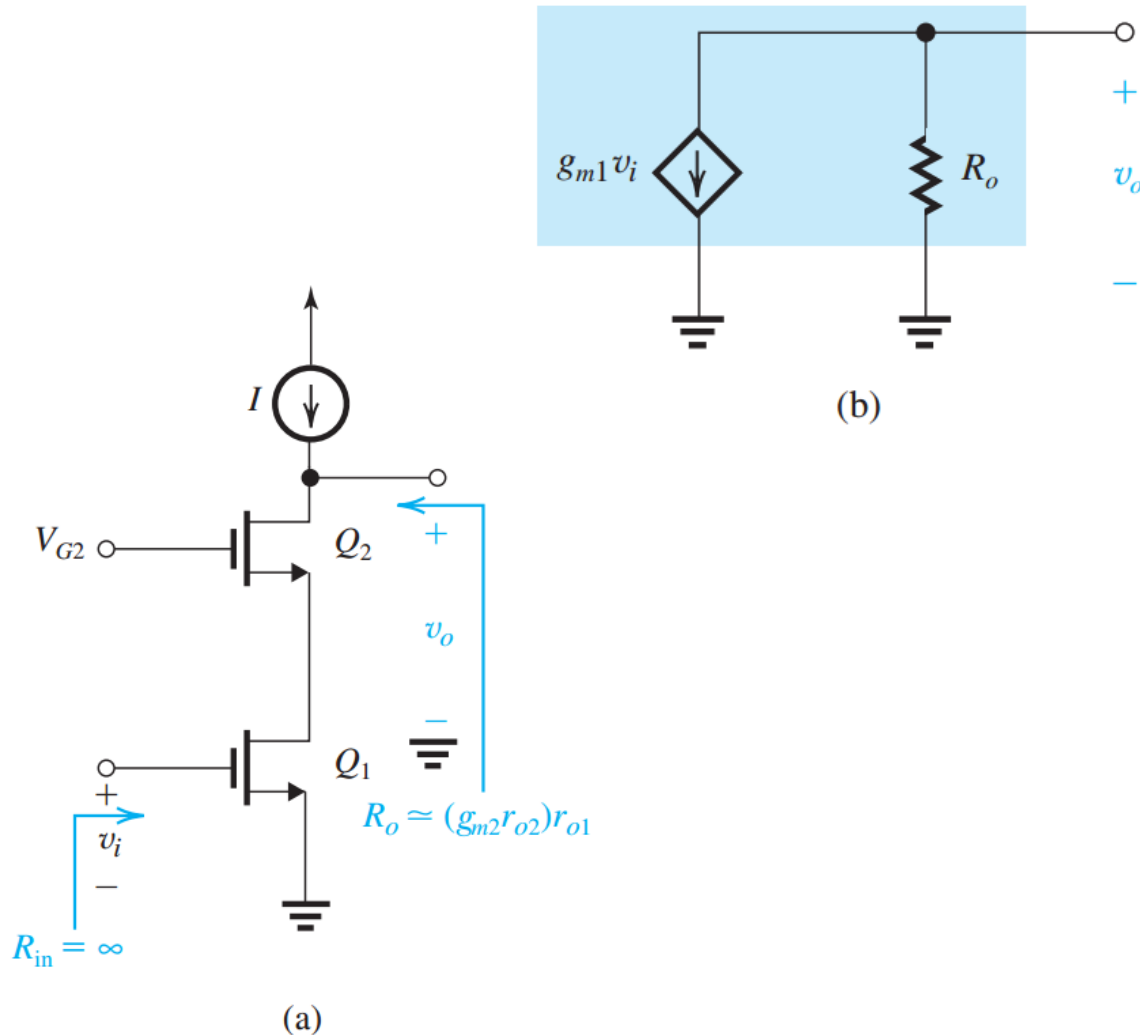
Continued 2



$$R_{\text{out}} = r_o + R_s + g_m r_o R_s \\ \simeq r_o + (g_m r_o) R_s$$

$$R_{\text{in}} = \frac{r_o + R_L}{1 + g_m r_o} \\ \simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

MOS Cascode Amplifier



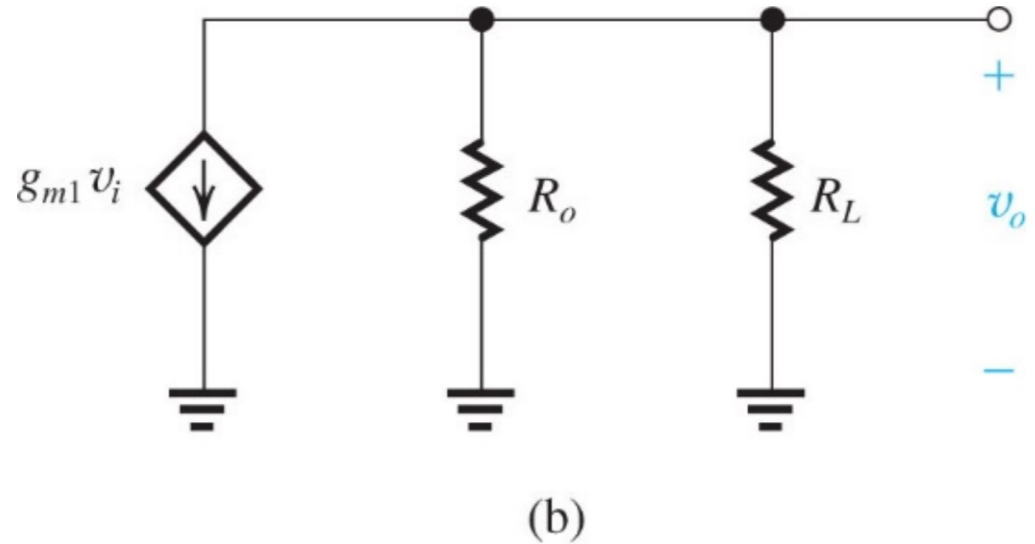
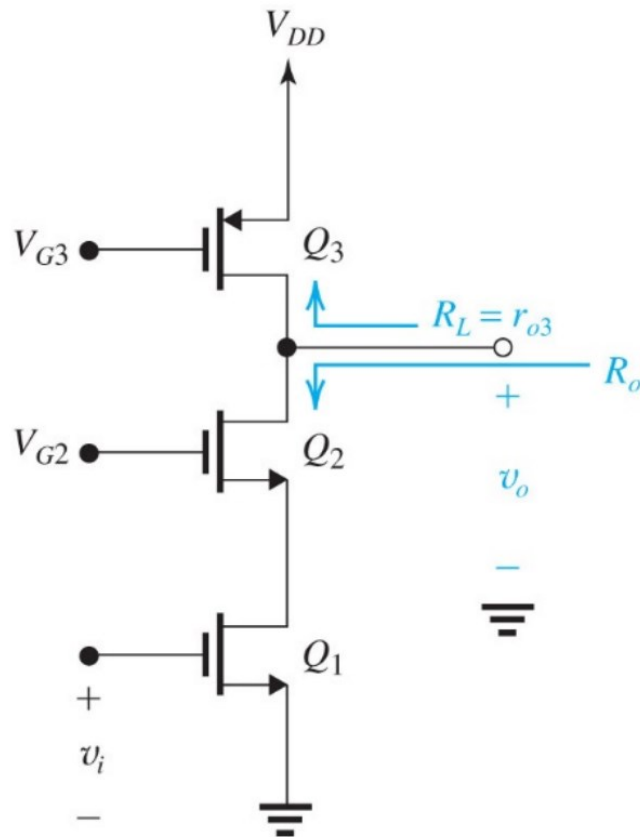
$$A_{vo} \equiv \frac{v_o}{v_i} = -g_{m1} R_o$$

$$R_o \simeq (g_{m2} r_{o2}) r_{o1}$$

$$A_{vo} = -(g_{m1} r_{o1})(g_{m2} r_{o2})$$

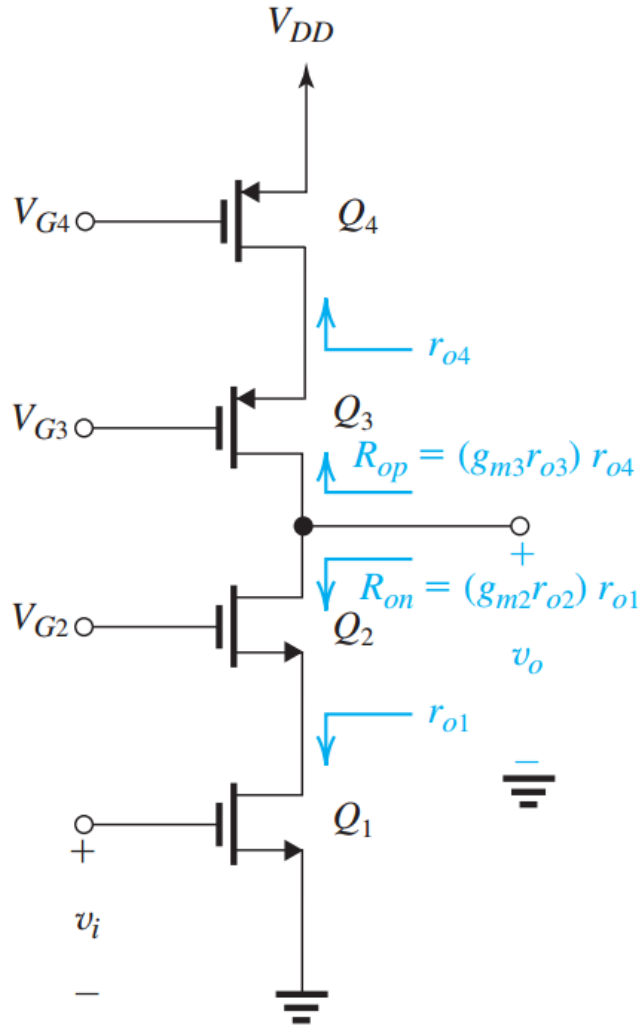
$$\begin{aligned} A_{vo} &= -(g_m r_o)^2 \\ &= -A_0^2 \end{aligned}$$

Cascode Amplifier with Simple Active Load

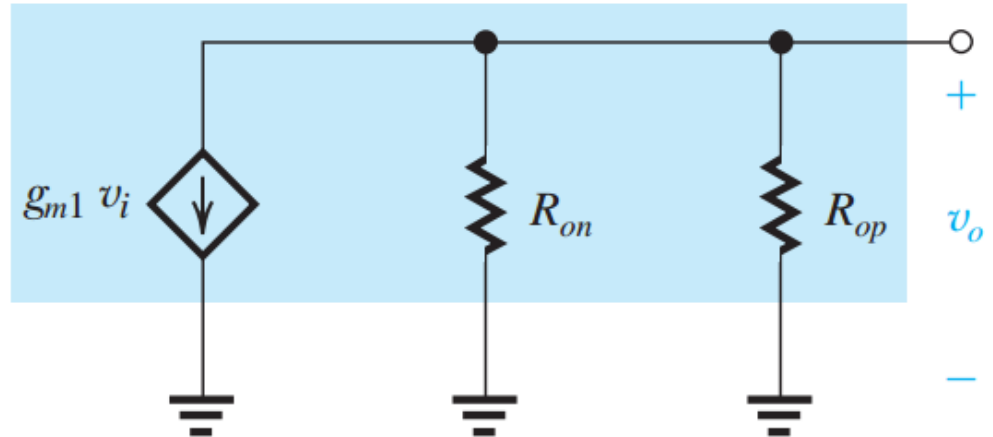


$$\begin{aligned}
 A_v &= -g_{m1}(R_o \parallel R_L) \\
 &= -g_{m1}(g_{m2}r_{o2}r_{o1} \parallel r_{o3}) \\
 &\simeq -g_{m1}r_{o3}
 \end{aligned}$$

Cascode Amplifier with Cascode Current-Source Load



(a)



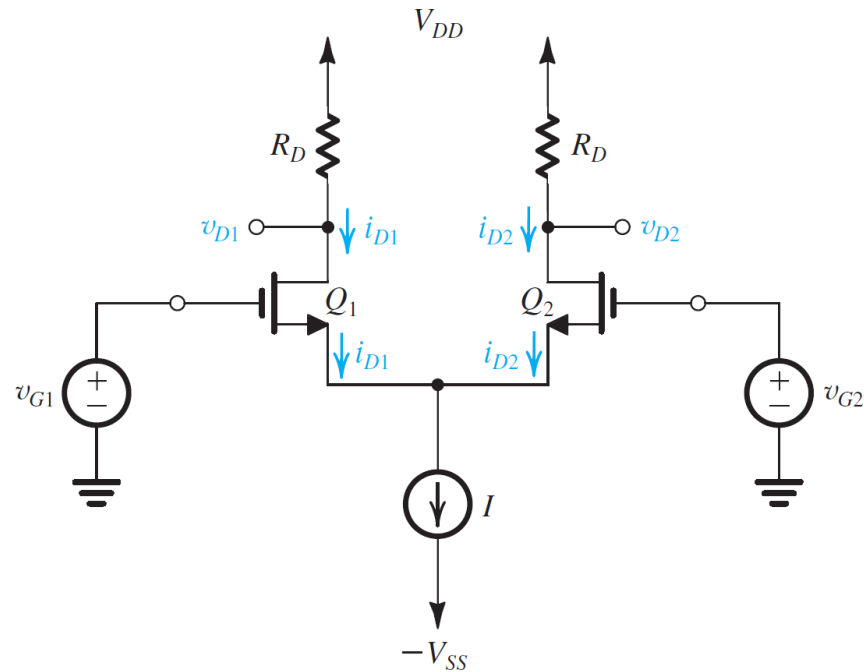
(b)

$$A_v = -g_{m1} \{ [(g_{m2} r_{o2}) r_{o1}] \parallel [(g_{m3} r_{o3}) r_{o4}] \}$$

$$A_v = -\frac{1}{2} (g_m r_o)^2 = -\frac{1}{2} A_0^2$$

MOS Differential-Pair

Basic Configuration



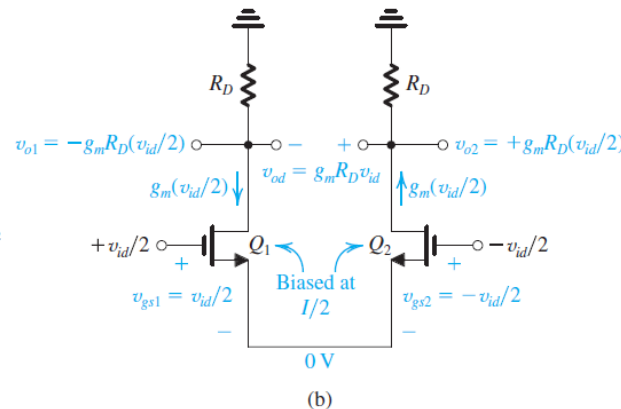
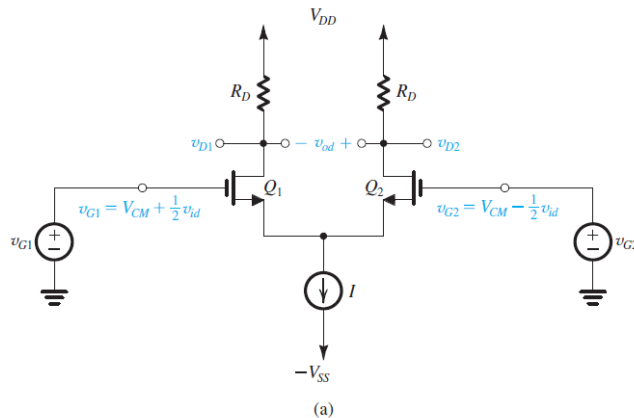
- Two matched MOS transistors
Common current bias.

“Differential signals” applied to v_{G1} and v_{G2} (equal amplitude but opposite sign).

“Differential outputs” are produced at v_{D1} and v_{D2} .

Note: in differential configuration with 0 differential input, V_{GS} is fixed for both Q_1 and Q_2 .

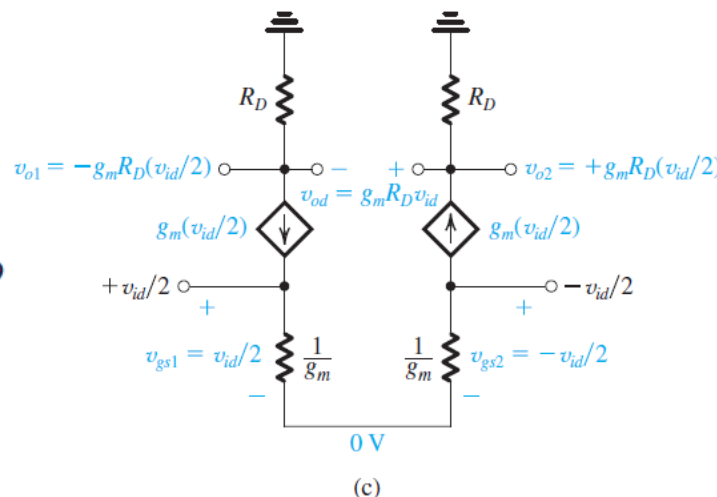
Small Signal Operation



For **differential AC small signal**, the differential pair is “anti-symmetric”. The potential at the mid point (Source) is zero. This is called “**Virtual Ground**”

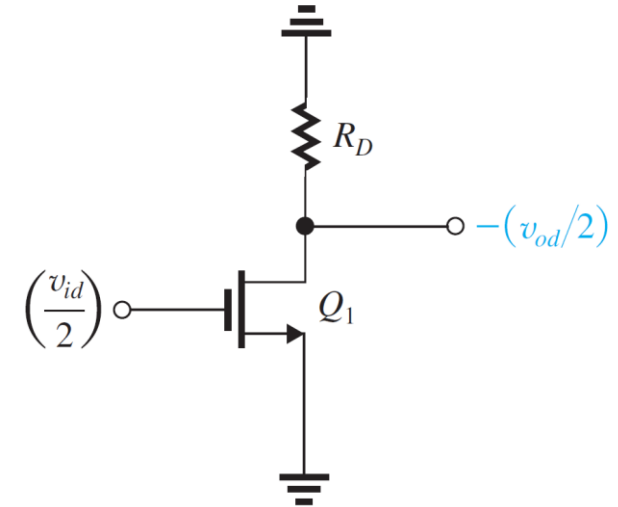
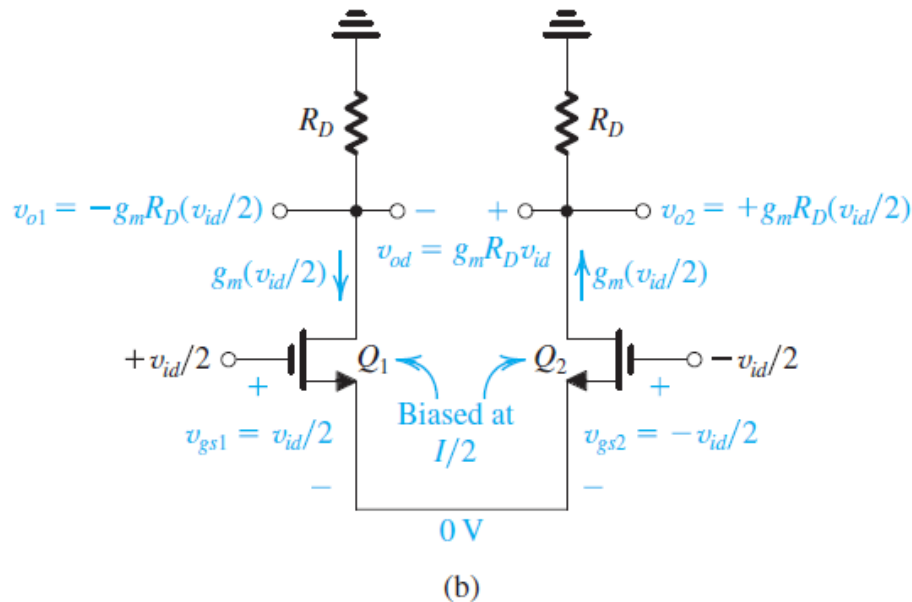
$$g_m = \frac{2I_D}{V_{OV}} = \frac{2(I/2)}{V_{OV}} = \frac{I}{V_{OV}}$$

$$A_d \equiv \frac{v_{od}}{v_{id}} = \frac{v_{o2} - v_{o1}}{v_{id}} = g_m R_D$$



This virtual ground is obtained without using a large bypass capacitor
-> much smaller area and better frequency response

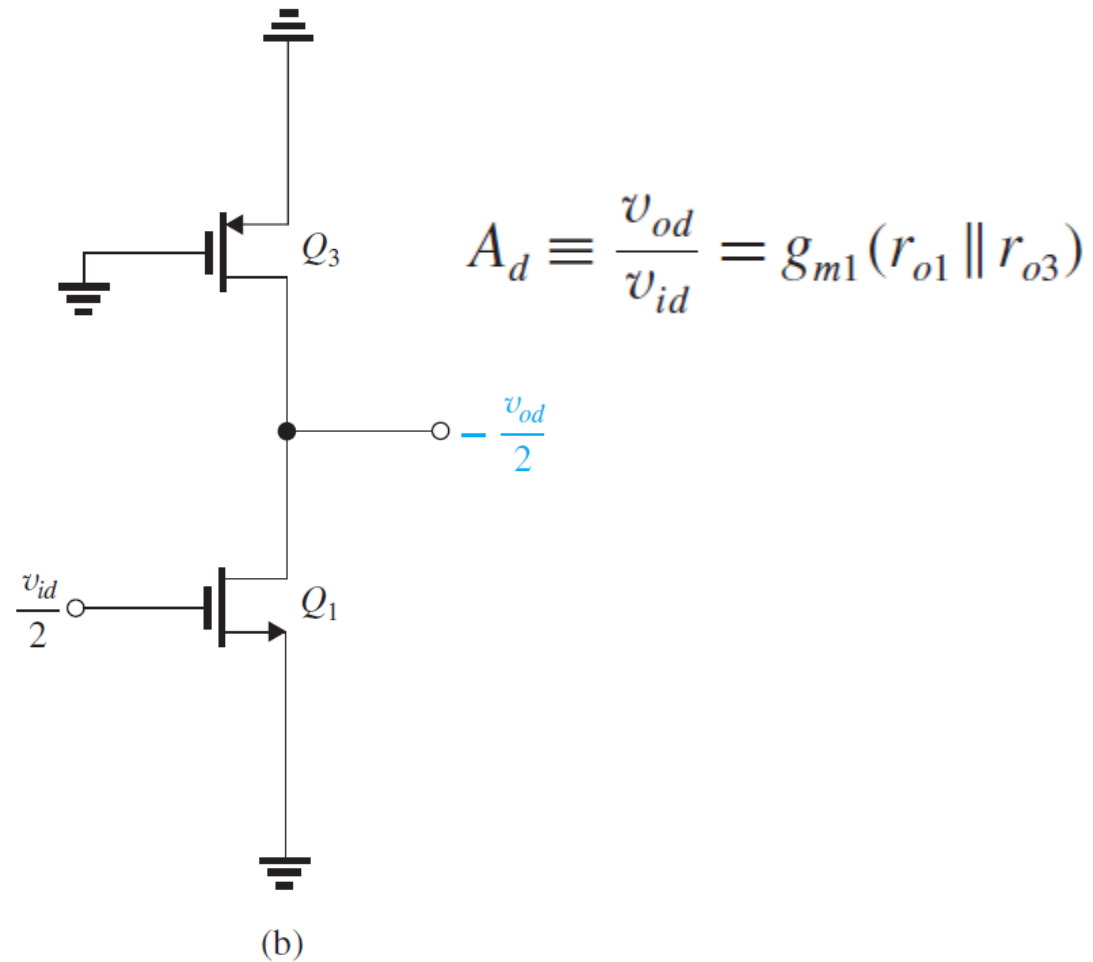
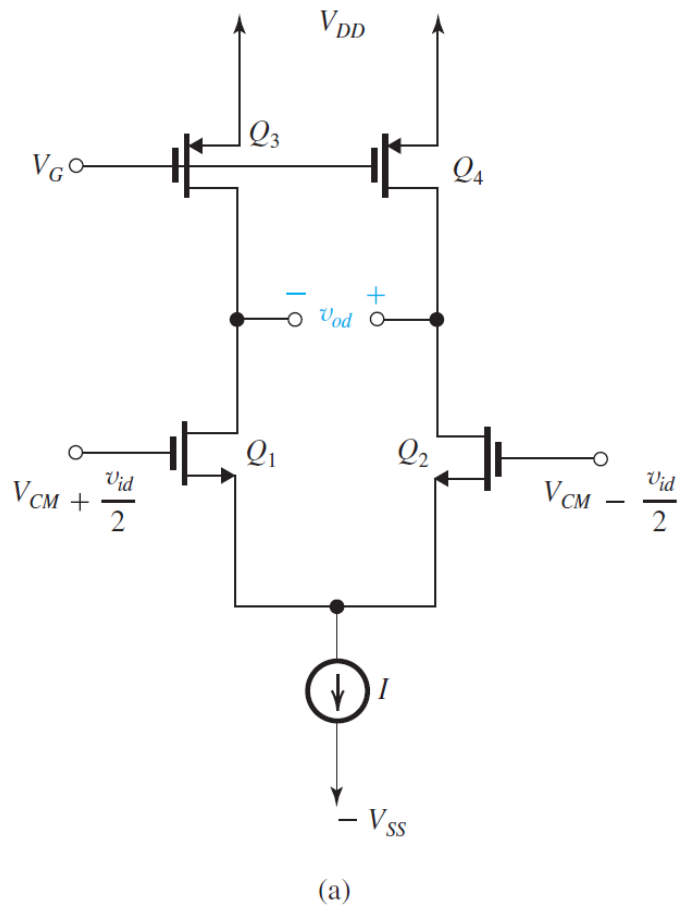
Differential Half Circuit



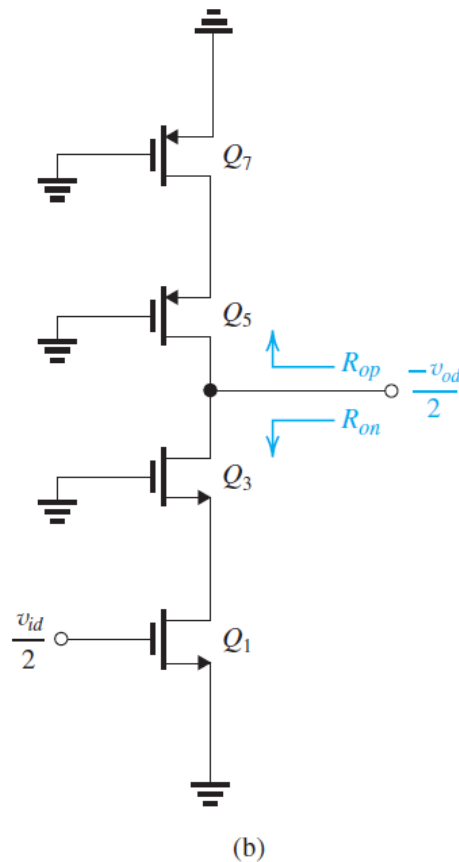
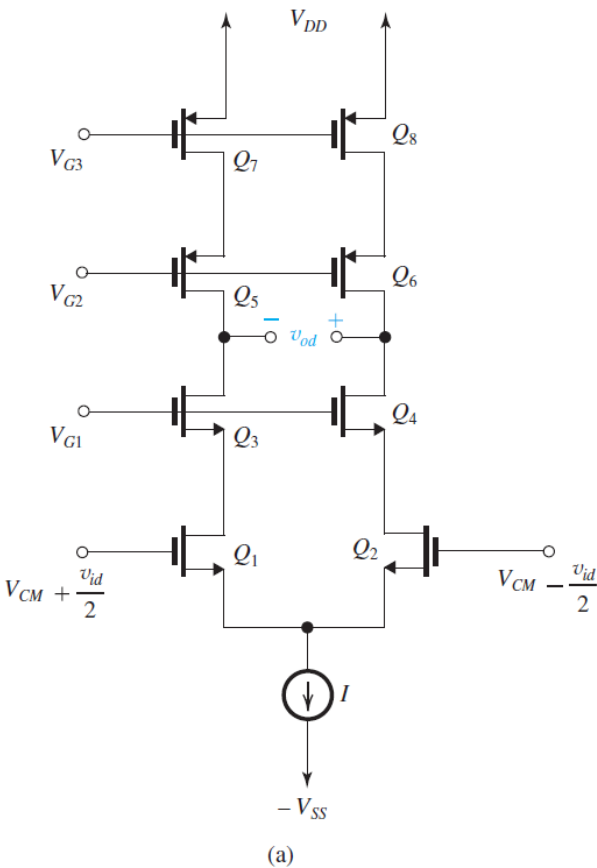
- Because the two halves of the circuits are anti-symmetric, and Source is at virtual ground, we can simplify and just analyze the "half circuit"

$$A_d = g_m (R_D \parallel r_o)$$

Differential Amplifier with Current-Source Loads



Cascode Differential Amplifier



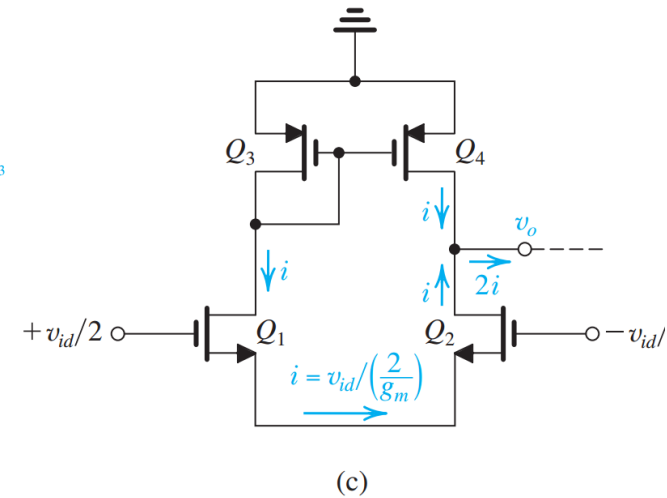
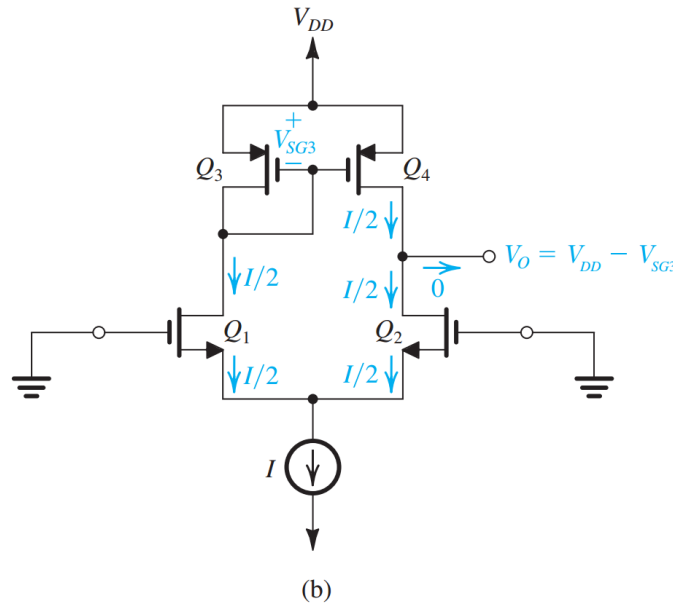
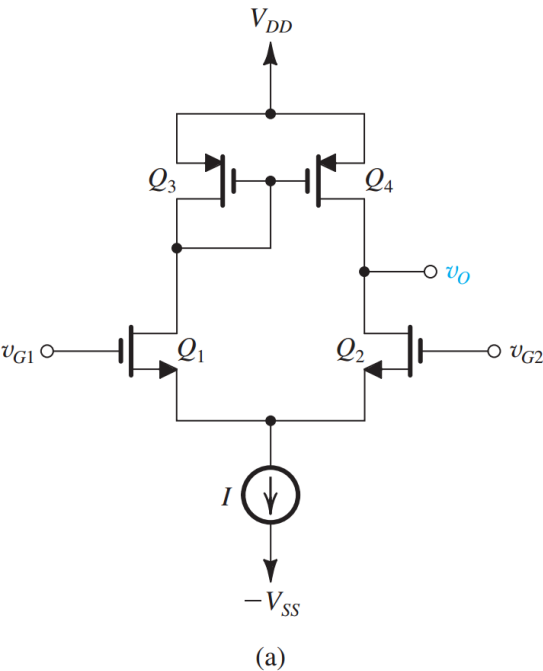
Cascode configurations for both amplifying transistors and current source loads.

$$A_d \equiv \frac{v_{od}}{v_{id}} = g_{m1} (R_{on} \parallel R_{op})$$

$$R_{on} = (g_{m3} r_{o3}) r_{o1}$$

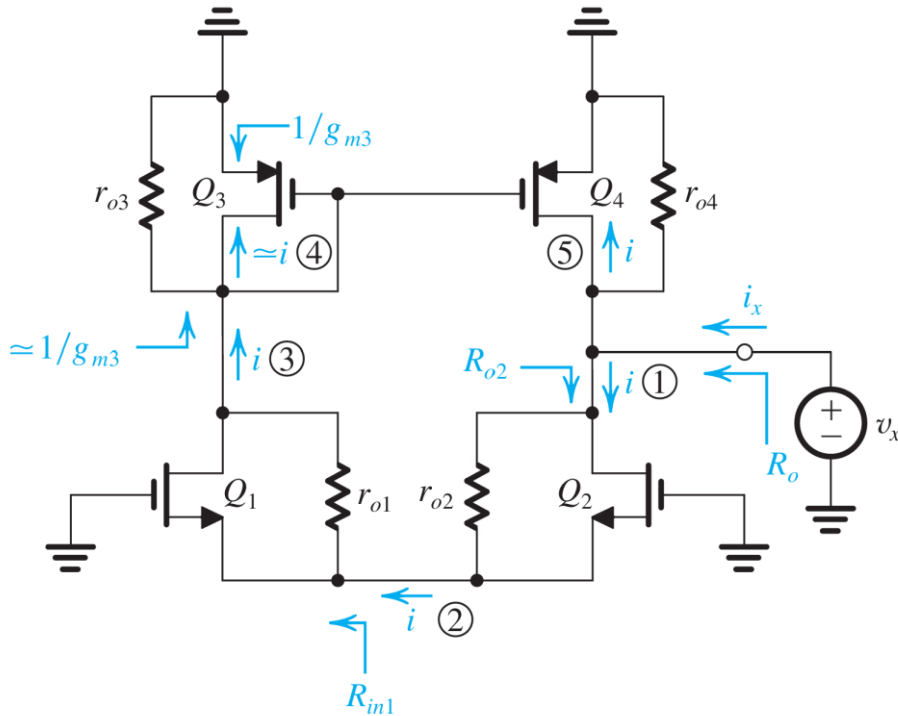
$$R_{op} = (g_{m5} r_{o5}) r_{o7}$$

MOS Differential Pair with Current Mirror Load



$$A_d \equiv \frac{v_o}{v_{id}} = G_m R_o = g_{m1,2} (r_{o2} \parallel r_{o4})$$

MOS Differential Pair with Current Mirror Load



$$\begin{aligned}
 R_{in1} &= \frac{r_{o1} + R_L}{g_{m1} r_{o1}} \\
 &= \frac{1}{g_{m1}} + \frac{1/g_{m3}}{g_{m1} r_{o1}} \simeq \frac{1}{g_{m1}}
 \end{aligned}$$

$$\begin{aligned}
 R_{o2} &= R_{in1} + r_{o2} + g_{m2} r_{o2} R_{in1} \\
 &= \frac{1}{g_{m1}} + r_{o2} + \left(\frac{g_{m2}}{g_{m1}} \right) r_{o2}
 \end{aligned}$$

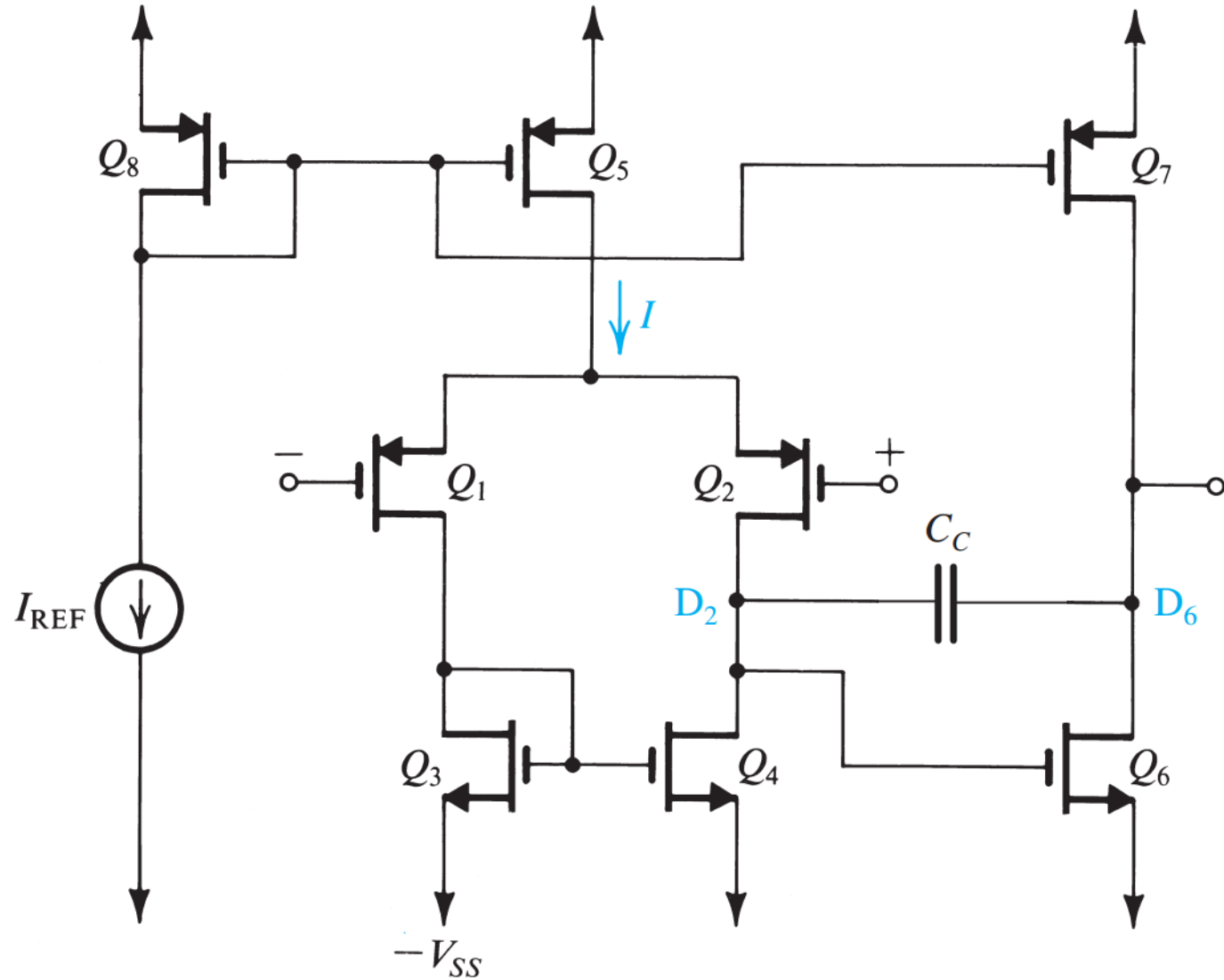
$$R_{o2} \simeq 2r_{o2}$$

$$\begin{aligned}
 i_x &= i + i + \frac{v_x}{r_{o4}} \\
 &= 2i + \frac{v_x}{r_{o4}} = 2 \frac{v_x}{R_{o2}} + \frac{v_x}{r_{o4}}
 \end{aligned}$$

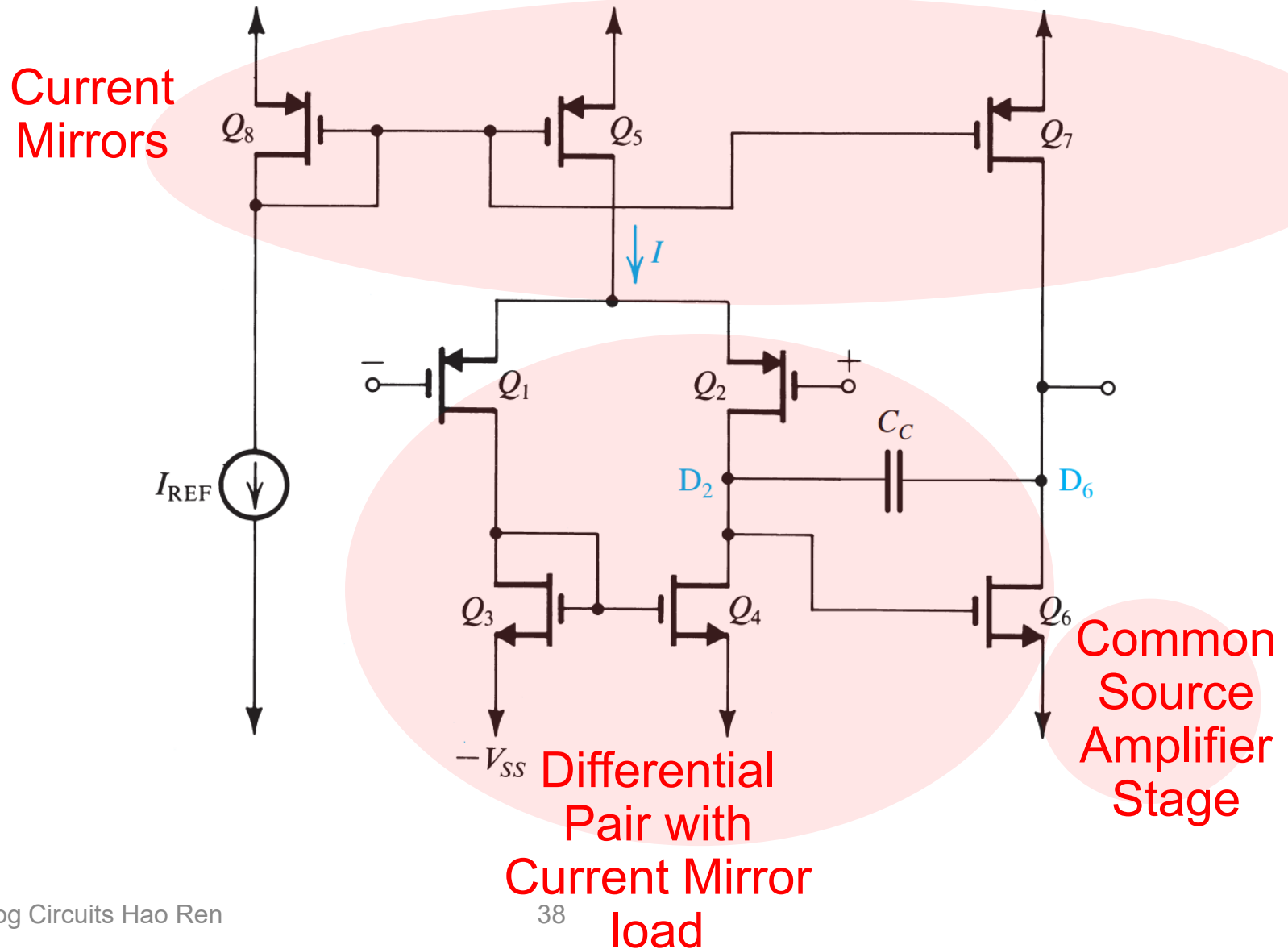
$$i_x = 2 \frac{v_x}{2r_{o2}} + \frac{v_x}{r_{o4}}$$

$$R_o \equiv \frac{v_x}{i_x} = r_{o2} \parallel r_{o4}$$

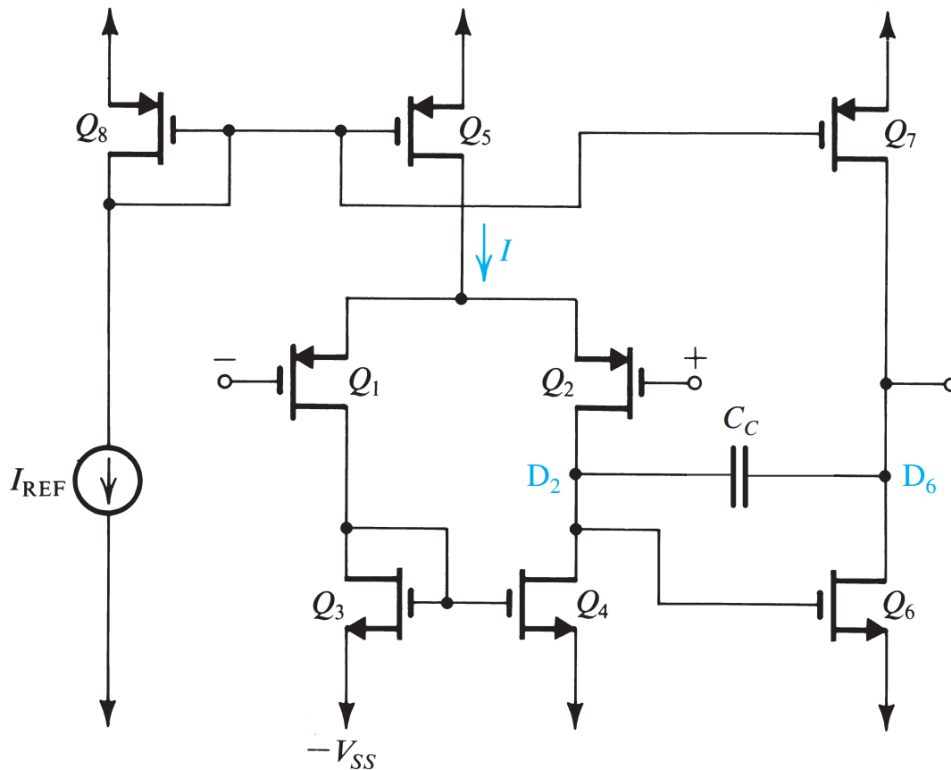
Two-Stage CMOS Op-Amp Circuit



Stages Scrutinize



Voltage Gains



- Voltage gain of the first stage (Q_1, Q_2): Differential input, single-ended output:

$$A_1 = -g_{m1}(r_{o2} \parallel r_{o4})$$

- Voltage gain of the 2nd stage (Q_6): Common source with current source load:

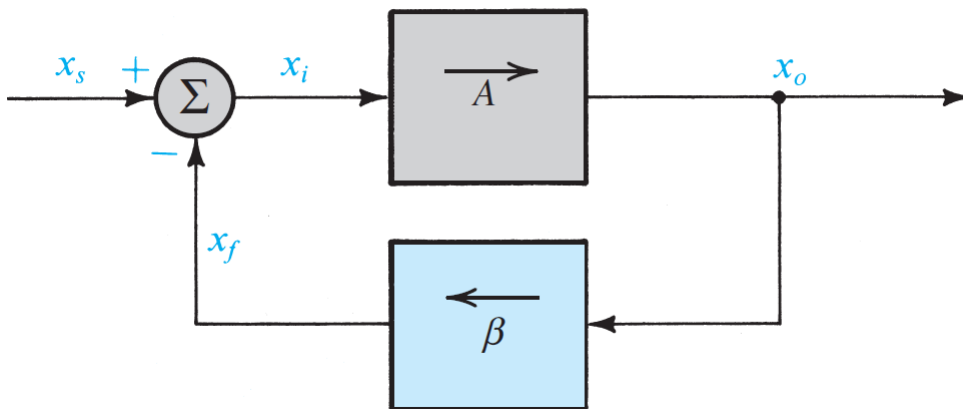
$$A_2 = -g_{m6}(r_{o6} \parallel r_{o7})$$

- Total gain

$$A_o = A_1 A_2$$

The general feedback structure

- A: open loop gain
- β : feedback factor



$$x_o = Ax_i$$

$$x_f = \beta x_o$$

$$x_i = x_s - x_f$$

$$A_f \equiv \frac{x_o}{x_s}$$

$$A_f = \frac{A}{1 + A\beta}$$

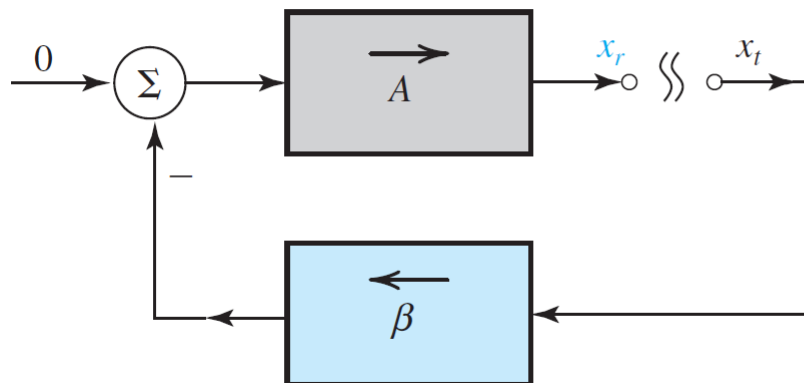
- $A\beta$: loop gain
- $A\beta \gg 1$

$$A_f \simeq \frac{1}{\beta}$$

The loop gain

- $A\beta$: loop gain
- $A\beta \gg 1$
- The sign of $A\beta$ determines the polarity of the feedback
- The magnitude of $A\beta$ determines how close the closed-loop gain A_f is to the ideal value of $1/\beta$.
- The magnitude of $A\beta$ determines the amount of feedback $(1+A\beta)$ the magnitude of the various improvements in amplifier performance

$$A_f \simeq \frac{1}{\beta}$$



$$x_r = -A\beta x_t$$

$$A\beta = -\frac{x_r}{x_t}$$

Figure 11.2 Determining the loop gain by breaking the feedback loop at the output of the basic amplifier, applying a test signal x_t , and measuring the returned signal x_r : $A\beta \equiv -x_r/x_t$.



Feedback gain desensitivity

$$A_f = \frac{A}{1 + A\beta}$$

$$dA_f = \frac{dA}{(1 + A\beta)^2}$$

$$\frac{dA_f}{A_f} = \frac{1}{(1 + A\beta)} \frac{dA}{A}$$

- the percentage change in A_f (due to variations in some circuit parameter) is smaller than the percentage change in A by a factor equal to the amount of feedback. For this reason, the amount of feedback, $1 + A\beta$, is also known as the desensitivity factor.

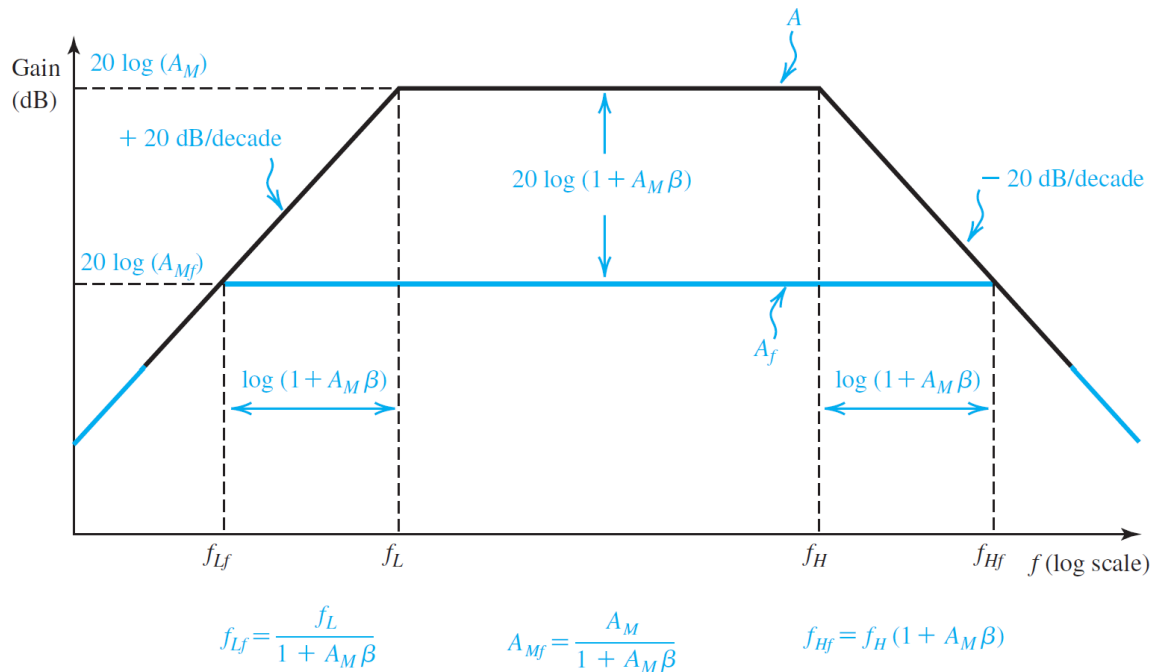
Feedback bandwidth extension

$$A(s) = \frac{A_M}{1 + s/\omega_H}$$

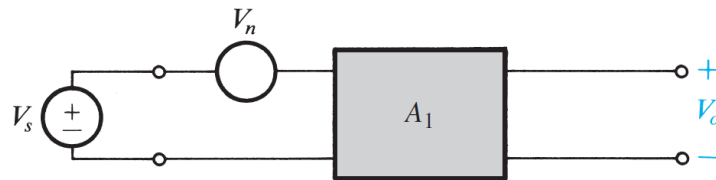
$$A_f(s) = \frac{A_M/(1 + A_M\beta)}{1 + s/\omega_H(1 + A_M\beta)}$$

$$A_f(s) = \frac{A(s)}{1 + \beta A(s)}$$

$$\omega_{Hf} = \omega_H(1 + A_M\beta)$$

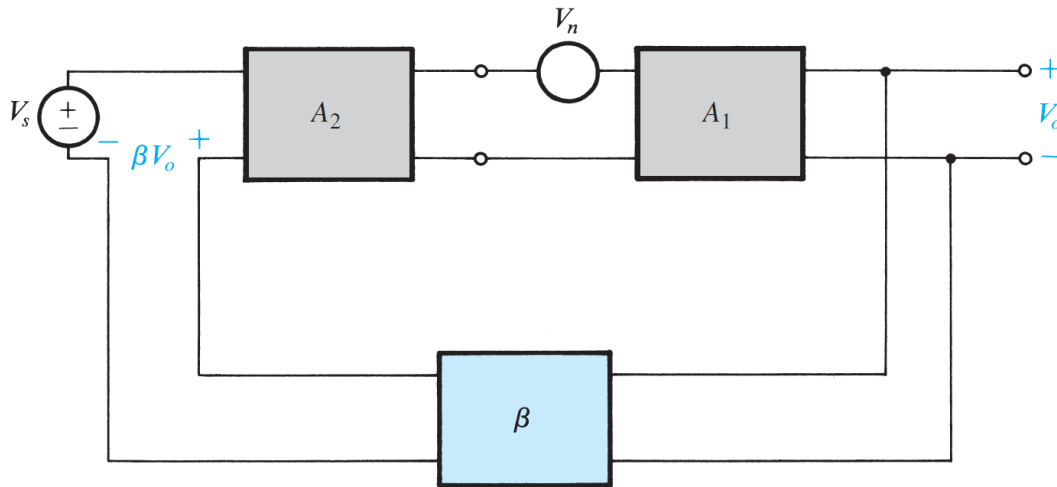


Feedback interference reduction



(a)

$$S/I = V_s/V_n$$



(b)

$$V_o = V_s \frac{A_1 A_2}{1 + A_1 A_2 \beta} + V_n \frac{A_1}{1 + A_1 A_2 \beta}$$

$$\frac{S}{I} = \frac{V_s}{V_n} A_2$$

Reduction in nonlinear distortion

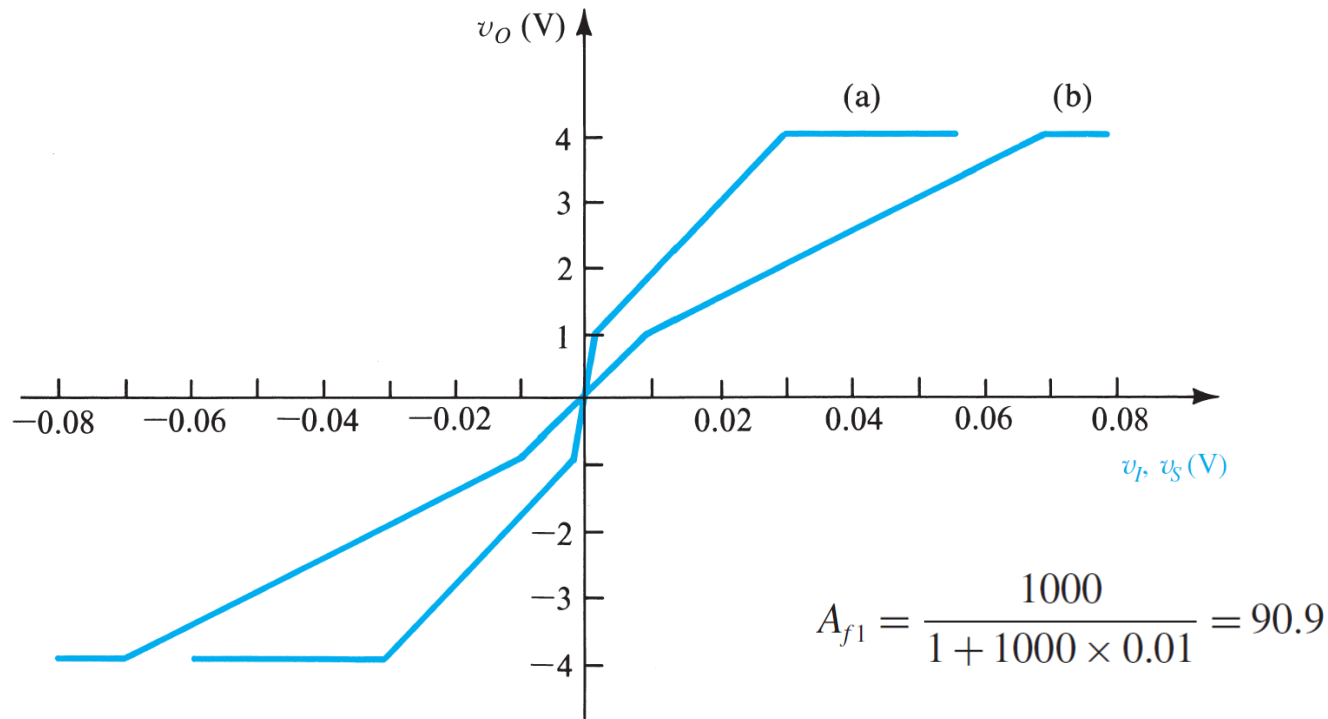


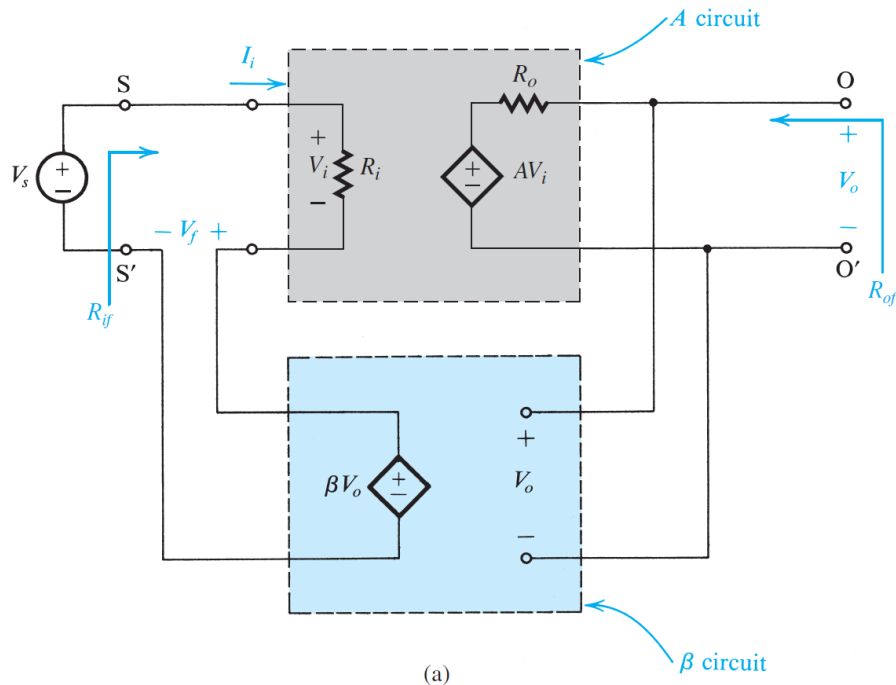
Figure 11.6 Illustrating the application of negative feedback to reduce the nonlinear distortion in amplifiers. Curve (a) shows the amplifier transfer characteristic (v_O versus v_i) without feedback. Curve (b) shows the characteristic (v_O versus v_s) with negative feedback ($\beta = 0.01$) applied.

$$A_{f2} = \frac{100}{1 + 100 \times 0.01} = 50$$

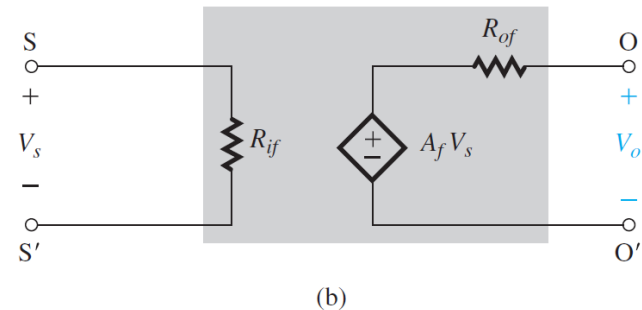
Systematic Analysis of Feedback Voltage Amplifiers



- The ideal structure of the series–shunt feedback amplifier consists of a unilateral open-loop amplifier (the A circuit) and an ideal voltage-sampling, voltage-mixing feedback network (the β circuit). The A circuit has an input resistance R_i , an open-circuit voltage gain A , and an output resistance R_o . It is assumed that the source is ideal with a zero resistance and that there is no load resistance. Also, no loading effect is considered.



$$A_f \equiv \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$$



The series–shunt feedback amplifier: (a) ideal structure; (b) equivalent circuit.

Systematic Analysis of Feedback Voltage Amplifiers

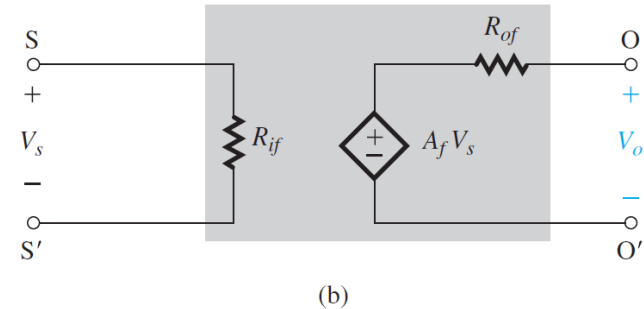
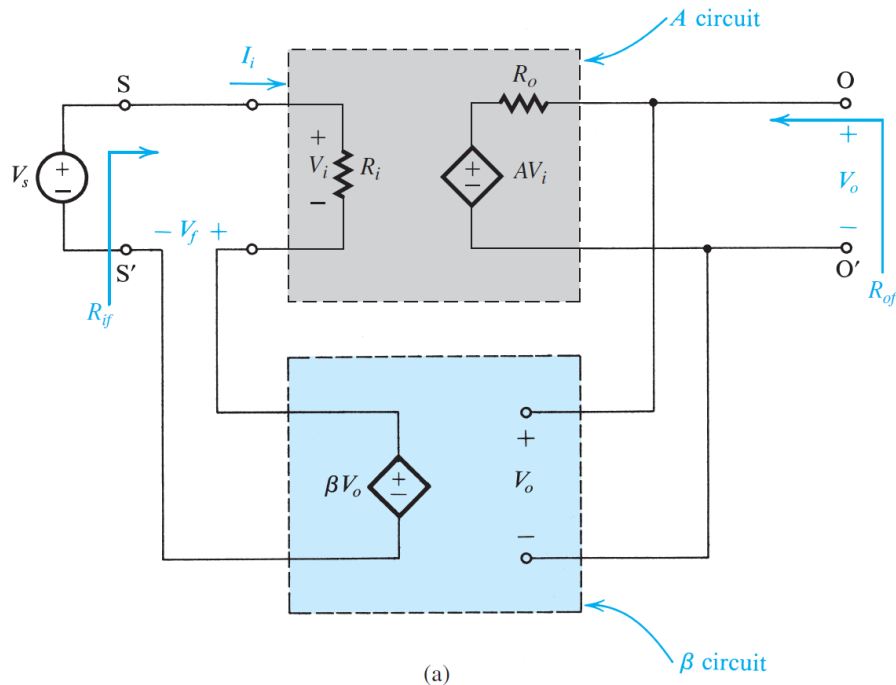


$$V_i = \frac{V_s}{1 + A\beta}$$

$$I_i = \frac{V_i}{R_i} = \frac{V_s}{(1 + A\beta)R_i}$$

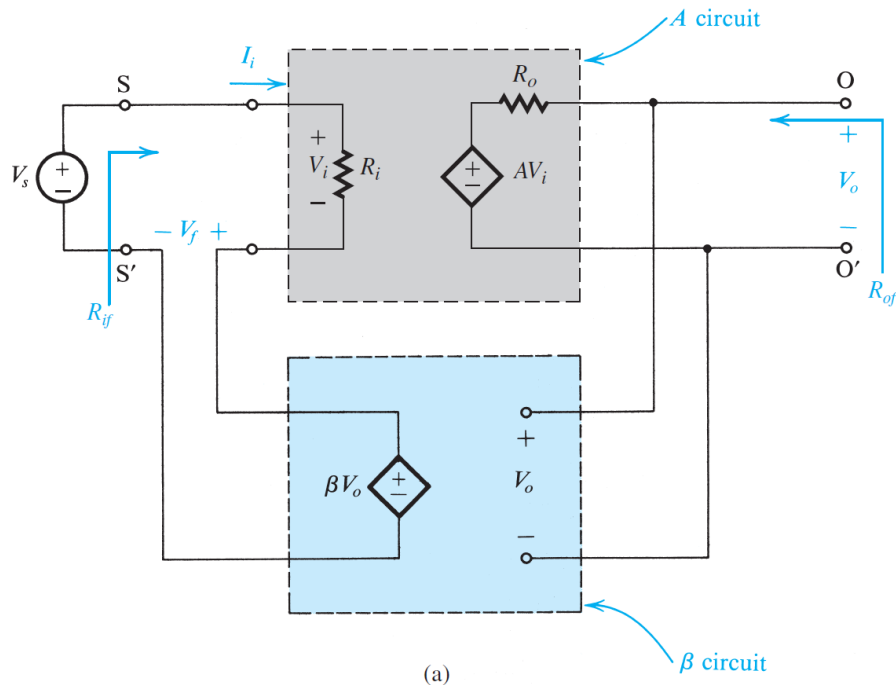
$$R_{if} \equiv \frac{V_s}{I_i}$$

$$R_{if} = (1 + A\beta)R_i$$



The series-shunt feedback amplifier: (a) ideal structure; (b) equivalent circuit.

Systematic Analysis of Feedback Voltage Amplifiers



$$R_{of} \equiv \frac{V_x}{I_x}$$

$$I_x = \frac{V_x - AV_i}{R_o}$$

$$V_i = -V_f$$

$$V_f = \beta V_o = \beta V_x$$

$$V_i = -\beta V_x \quad \Rightarrow \quad R_{of} = \frac{R_o}{1 + A\beta}$$

$$I_x = \frac{V_x(1 + A\beta)}{R_o}$$

